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Bovine-LTID: A vision-based spatio-temporal identity system for multi-camera cattle tracking and long-term behavior monitoring

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ABSTRACT

Multi-camera video monitoring of group-housed cattle is essential for health assessment in precision livestock farming, offering a scalable, non-invasive alternative to labor-intensive manual observations and restrictive wearable sensors. However, existing systems for cross-camera identity matching typically rely on computationally intensive appearance-based Re-identification (ReID). These approaches often struggle in dense barns due to frequent mutual occlusions, viewpoint variations, and the high visual similarity of cattle coat patterns. To address these limitations, we propose Bovine Long-Term Identity (Bovine-LTID), a resource-efficient, ReID-embedding-free spatio-temporal tracking framework for multi-camera behavior monitoring. Instead of relying on ambiguous appearance features, Bovine-LTID maintains identity consistency through spatial geometry and temporal logic. The system integrates a behavior-aware detector (YOLOv9s) with a cattle-optimized single-view tracker (CowTracker) to recover identities during prolonged occlusions. For cross-view synchronization, trajectories are projected into a unified ground-plane coordinate system via homography calibration and managed by a centralized Global ID Pool. Trained on NVIDIA H200 GPU, the detection module achieves a test-set mAP50 of 0.944, while the complete pipeline operates at approximately 13 FPS on consumer-grade hardware (NVIDIA RTX 4060 Laptop, 8 GB VRAM). The cross-camera fusion mechanism reduces global raw identity switches by 87.6% over single-camera baselines and achieves a 99.33% cross-view identity consistency. These findings present a practical methodology for scalable, automated herd welfare assessment spanning from high-performance training to edge deployment.

1. Introduction

The continuous monitoring of individual cattle identity and behavior is a key component of modern precision livestock farming, supporting herd welfare, early disease detection, and overall productivity. Traditionally, farm management has relied on manual observation by experienced stockpersons (Li et al., 2026). While human observation provides holistic welfare assessments, it is labor-intensive, subject to human variance, and difficult to scale for continuous, 24/7 monitoring in commercial herds.

To automate monitoring, the industry subsequently transitioned to contact-based and wearable sensor technologies. These solutions encompass various devices, including Radio Frequency Identification (RFID) tags for localization and identification (Peng et al., 2025), neck-mounted Inertial Measurement Units (IMUs) and accelerometers for

kinetic behavior analysis (e.g., resting, walking, rumination) (Haladjian et al., 2018; Lovarelli et al., 2022), and pedometers or ingestible rumen boluses for physiological monitoring (Hajnal et al., 2022). While these physical devices provide localized measurements, they present operational limitations in large-scale applications. Specifically, they require battery and hardware maintenance, incur per-animal deployment costs, and involve physical attachment that can induce stress or discomfort, potentially affecting animal welfare (Rao et al., 2025). Furthermore, isolated single-point sensor data may lack the contextual information needed to differentiate complex social or spatial behavioral states.

Non-invasive camera-based monitoring offers a promising approach to mitigate these contextual and operational limitations (Wang et al., 2021; Antognoli et al., 2025), yet the long-term tracking reliability of initial single-camera configurations remains inherently limited by

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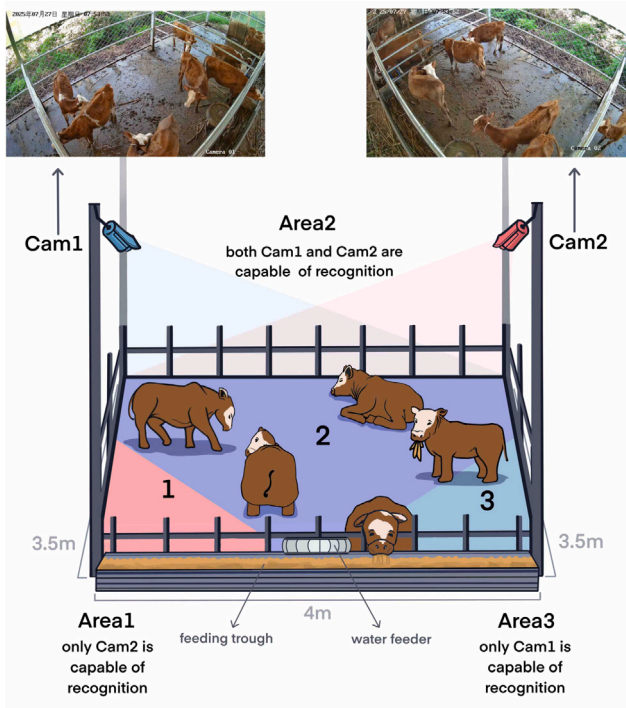


Fig. 1. Conceptual overview of the experimental setup and monitoring scenario. Relying on a single viewpoint inevitably leads to severe occlusions in dense barns. Bovine-LTID mitigates this by utilizing strategically deployed overlapping camera views to ensure continuous spatio-temporal tracking.

physical constraints. Driven by advancements in computer vision and deep learning, various iterations of the YOLO architecture and Convolutional Neural Networks (CNNs) have been extensively adapted for real-time cattle behavior classification, such as feeding, ruminating, and drinking (Rao et al., 2025; Tong et al., 2024; Jiang et al., 2023). Concurrently, multi-object trackers and pose-estimation frameworks have been employed to automate individual herd observation and skeletal tracking (Myat Noe et al., 2025; Li et al., 2022; Islam et al., 2023). Some studies, such as the Cows2021 dataset (Gao et al., 2021, 2022), explored video-based cattle re-identification by leveraging temporal cues, while other advanced approaches have incorporated depth data for deep metric learning (Sharma et al., 2025), or utilized architectures like HanwooReID (Liu et al., 2025) for pose-aware feature learning. Despite being straightforward to deploy, these single-camera setups are susceptible to dense occlusion (particularly in crowded feeding troughs), restricted fields of view, and variable illumination. As highlighted by recent comprehensive reviews on livestock multi-object tracking (MOT), navigating these frequent long-term occlusions, visually homogeneous animal appearances, and complex non-linear motion patterns remain the primary bottlenecks for continuous single-view welfare monitoring (Nidhi et al., 2025b).

To address these limitations, multi-camera systems offer complementary perspectives and expanded spatial coverage. Recent large-scale benchmarks, notably MultiCamCows (Yu et al., 2025), have shown that multi-view configurations improve recognition robustness. Recent state-of-the-art methodologies have also explored utilizing skeletal keypoints to track cattle across multiple cameras (Nidhi et al., 2025a), as well as employing Bird's-Eye-View (BEV) projections to associate trajectories across overlapping fields of view (Han et al., 2025). However, maintaining consistent identity synchronization across views remains a technical challenge. Many cross-camera frameworks still depend heavily on appearance-based Re-identification (ReID) or periodic localized facial recognition models to correct identity drift (Han et al., 2025; Xu

et al., 2021). Other recent pipelines utilize deep visual feature extractors (e.g., VGG) coupled with classifiers (e.g., SVM) for individual cattle identification (Mon et al., 2024). Applying these appearance-based methods to commercial cattle herds introduces practical challenges: first, cattle of the same breed (e.g., Simmental) exhibit high visual homogeneity and similar coat patterns, making visual matching inherently brittle; second, extracting high-dimensional appearance features or isolating frontal facial crops for every instance across multiple frames incurs considerable computational overhead, limiting real-time deployment on standard farm edge devices. This highlights the need for a more efficient, scalable, and ReID-embedding-free approach.

To address these gaps, we introduce Bovine-LTID, a vision-based spatio-temporal identity framework designed for multi-camera cattle tracking and long-term behavior analytics in small-group indoor barn environments.

The system leverages a behavior-aware detection model (YOLOv9s (Wang et al., 2025)) integrated with CowTracker, an enhanced ByteTrack-based (Zhang et al., 2022) module adapted for livestock, to maintain tracking persistence under occlusion without relying on appearance feature extraction.

As conceptually illustrated in Fig. 1, the experimental design of Bovine-LTID utilizes overlapping viewpoints from multiple cameras to mitigate the blind spots inherent to single-camera systems. To stabilize tracking within each individual view, CowTracker incorporates multi-constraint target association and identity stability protection, augmented by spatial mask refinement (Zhou et al., 2023). At the cross-camera level, the system projects disparate camera views into a unified ground-plane coordinate space via homography calibration. A centralized Global ID Pool (GIDP) is then employed to maintain identity synchronization across overlapping regions based on spatio-temporal logic, avoiding reliance on appearance-based ReID. Furthermore, a cross-view behavior inference engine uses multiple viewpoints to refine uncertain behavior labels, assisting in resolving identity fragmentation and facilitating long-term behavioral analytics (e.g., individual time-budget estimation).

The primary contributions of this work are summarized as follows:

- We propose Bovine-LTID, a resource-efficient, ReID-embedding-free multi-camera tracking framework. By utilizing spatio-temporal consistency rather than computationally intensive appearance features, the system addresses the challenge of tracking visually similar cattle in complex farm environments.
- We develop CowTracker, a cattle-specific tracking module built upon the ByteTrack architecture. By incorporating multi-constraint target association and spatial mask refinement, this method improves trajectory continuity and occlusion recovery under dense housing conditions.
- We introduce a homography-based Global ID Pool and a cross-view behavior inference mechanism to mitigate identity fragmentation across overlapping camera views. This integration supports continuous, individual-level behavioral analytics in real-time on consumer-grade hardware, offering a practical solution for automated welfare monitoring at a small-group scale.

2. Data acquisition and system architecture

This section presents the data collection process and implementation details of the Bovine-LTID system, including behavior-aware detection, camera calibration and region initialization, single-camera tracking, and multi-camera identity fusion.

2.1. Data acquisition and dataset construction

Video data were collected in a commercial beef cattle barn in China using two ceiling-mounted Hikvision 4K surveillance cameras, recording at a native frame rate of 10 frames per second (FPS). To

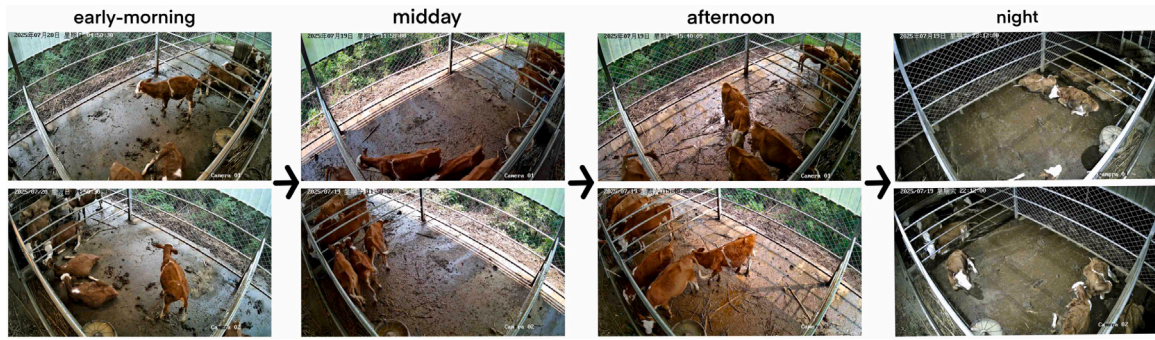


Fig. 2. Representative frames illustrating illumination variation across different times of day (from left to right): early morning, midday, afternoon, and night. These conditions highlight the visual complexity of real-world dairy barn monitoring.



Fig. 3. Representative examples of annotated cattle behaviors in the training dataset (left to right): *Stand*, *Unclear*, *Lying down*, *Drinking water*, and *Foraging*. Each instance is labeled with a bounding box and behavior class to support joint detection and classification.

Table 1

Dataset statistics for YOLOv9s training.

Split	Frames	Bounding boxes
Train	2741	6050
Val	783	1736
Test	393	875
Total	3917	8661

ensure robust generalization across diverse spatial densities and environmental conditions, the data collection encompasses intermittent full-day video recordings covering variations of 1, 3, and 5 cattle within the monitoring zone. These multi-period recordings capture a wide spectrum of illumination contexts, including early morning, midday, afternoon, and nighttime under artificial lighting. Each camera covered the feeding and resting zones with partial field-of-view overlap to maintain continuous monitoring of Simmental cattle activities.

To illustrate the lighting variability in the barn across the day, representative frames from different periods are shown in Fig. 2. These illumination changes – ranging from natural daylight to low-light artificial conditions – introduce significant visual complexity and challenge stable detection and tracking, underscoring the need for a high-quality, real-world dataset.

Frames were uniformly sampled from the recorded videos and manually annotated using the open-source graphical annotation tool, LabelImg. Each cow instance was carefully labeled with a bounding box and classified into one of five behavior categories: *standing*, *lying*, *foraging*, *drinking*, or *unclear*. Fig. 3 shows representative annotation samples for each behavior category.

In total, 3917 frames were annotated, yielding 8661 cattle instances across the five behavior categories. The dataset was split into training, validation, and test sets as summarized in Table 1.

Scientific rationale for the “unclear” category. While the inclusion of an “unclear” behavior class may initially appear as a methodological concession, it is a scientifically necessary representation of the inherent visual ambiguity in single-view monitoring. Due to fixed camera

perspectives and inevitable blind spots, a cow may frequently be only partially visible in a given view. When critical anatomical features (e.g., the head or legs) are cut off by the viewing angle, specific behaviors become objectively indistinguishable from that single perspective alone. Rather than forcing the detection model to make overconfident, erroneous classifications – which would directly confound downstream behavioral analytics – the “unclear” class serves as an uncertainty filter.

During our real-world continuous monitoring sessions, single-camera behavior classification exhibited “unclear” detection rates of 21.4% (15,377 out of 71,924 detections for Camera 1) and 13.2% (9511 out of 71,786 detections for Camera 2). Crucially, this single-view ambiguity does not propagate to the final analytics. Through the cross-view behavior inference mechanism (detailed in Section 3.3), the system leverages complementary perspectives: it transfers the confident behavior label from the camera with a clear view to resolve the blind-spot ambiguity in the other. This cooperative fusion reduces the aggregate “unclear” rate to just 0.2% (only 145 out of 71,718 globally fused detections)—a reduction of over 98.5%. This indicates that the “unclear” category arises primarily from the inherent physical limitations of single-view observations rather than from shortcomings of the model itself. These limitations are effectively addressed by our multi-camera fusion architecture.

2.2. YOLO training and deployment configuration

To ensure a fair comparison, all evaluated YOLO variants were fine-tuned using an identical protocol. Training was conducted on a single NVIDIA H200 GPU (operating within an 8× H200 cluster for parallel processing) using mixed-precision training (AMP) to optimize throughput and stability. To enhance generalization under barn conditions (variable lighting, dust, and occlusion), a data-augmentation strategy was applied, consisting of rotation, translation, scaling, color jitter, Mosaic, and MixUp. Shear and perspective transforms were disabled to prevent geometrically implausible cattle postures that could mislead the classifier. As is standard practice, Mosaic augmentation was automatically deactivated during the last training stage to promote convergence and reduce overfitting risk.

Table 2

Detection performance of the fine-tuned YOLOv9s (Detector for Bovine-LTID).

Model	P (%)	R (%)	mAP50	mAP50-95
YOLOv9s	96.1 ± 0.3	94.0 ± 0.7	0.944 ± 0.003	0.885 ± 0.008

Training was initialized from official COCO-pretrained weights using the AdamW optimizer (Loshchilov and Hutter, 2019) with a cosine learning-rate schedule. To facilitate reproducibility, a summary of the shared training hyperparameters—including optimization variables, batch configuration, and augmentation bounds—is detailed in Table 3. During training, disk-based dataset caching and optimized data loading were applied to reduce I/O bottlenecks on the single-GPU setup, and early stopping based on validation mAP50 was used to determine the best checkpoint for each model.

Each detection output from the finally selected YOLOv9s model is represented as a tuple $D_i = \{b_i, c_i, s_i\}$, where $b_i \in \mathbb{R}^4$ denotes the normalized bounding-box coordinates, $c_i \in \{0, 1, 2, 3, 4\}$ represents the predicted behavior class (standing, lying down, foraging, drinking water, or unclear), and $s_i \in [0, 1]$ is the corresponding confidence score.

To satisfy real-time requirements on edge hardware while maintaining instance segmentation, EdgeSAM is executed periodically (Zhou et al., 2023) rather than on every frame. By default, EdgeSAM processes one out of every three frames, producing a refined binary mask m_i exclusively for detections with $s_i \geq 0.5$. To empirically justify this default configuration, a controlled sensitivity experiment was conducted testing intervals ranging from 1 to 10 frames. As detailed in Appendix C, the every-3-frames setting was selected as it delivers an approximately two-fold increase in system throughput (achieving ~13.6 FPS) with zero degradation to the global tracking metrics. The augmented detection output is therefore defined as:

$$D_i^{\text{aug}} = \begin{cases} \{b_i, m_i, c_i, s_i\}, & \text{if EdgeSAM is triggered on frame } i, \\ \{b_i, c_i, s_i\}, & \text{otherwise.} \end{cases} \quad (1)$$

This adaptive strategy reduces average computational cost while preserving accurate masks on the majority of frames. To clarify the operational speed of the system, it is necessary to distinguish between isolated module benchmarking and practical system deployment. While the standalone YOLOv9s model achieves a raw inference speed of 57.9 FPS (used for variant benchmarking in Section 4.1), the complete dual-camera Bovine-LTID pipeline operates at approximately 13 FPS on a system equipped with an Intel i9-14900HX CPU and an NVIDIA RTX 4060 Laptop GPU. This end-to-end workload accounts for simultaneous dual-stream decoding at native 1920×1080 resolution, YOLOv9s inference on both views, periodic EdgeSAM execution, CowTracker association logic, and cross-camera homography fusion.

Under varying illumination and mutual occlusions, the fine-tuned YOLOv9s achieves the detection performance summarized in Table 2.

Rather than optimizing for a single peak metric, YOLOv9s was selected for delivering a balance across Precision (96.1%), Recall (94.0%), overall mAP (0.944), and computational efficiency. This multi-metric performance provides a suitable foundation for downstream individual identification and behavior analysis in small-group indoor barn environments.

2.3. System architecture: Enabling real-time multi-view analytics

Implementing multi-camera, behavior-aware tracking in real-world agricultural environments poses computational challenges. The system must simultaneously process high-definition video streams, execute deep learning inferences (YOLOv9s and EdgeSAM), and perform complex spatio-temporal matching without dropping frames, which would otherwise disrupt behavioral trajectory continuity. To bridge the gap between computational requirements and limited edge resources,

Table 3

Shared training configuration and hyperparameters applied to all evaluated YOLO variants.

Parameter	Value
<i>Optimization</i>	
Optimizer	AdamW
Initial Learning Rate (lr0)	0.001
Final LR Fraction (lrf)	0.05 (Cosine Annealing)
Weight Decay	5×10^{-4}
Warmup Epochs	3
<i>Training Setup</i>	
Epochs	200 (Early stopping patience = 40)
Batch Size	32
Input Resolution	640×640 pixels
Mixed Precision (AMP)	True
<i>Data Augmentation</i>	
Mosaic/MixUp	1.0/0.10
HSV Perturbation (H, S, V)	0.015, 0.5, 0.4
Geometric Transforms	Degrees: 5°, Translate: 0.1, Scale: 0.4
Horizontal Flip (fliplr)	0.5

Bovine-LTID employs a concurrent, multi-threaded architecture (illustrated in Fig. 4). This architecture is designed to serve the core methodological objectives: strict temporal synchronization for cross-camera fusion, adaptive resource allocation for continuous tracking, and non-blocking behavioral analytics.

2.3.1. Synchronized dual-stream ingestion

Accurate cross-camera identity matching via homography projection (detailed in Section 3.3) inherently relies on precise temporal alignment between the overlapping views. To achieve this, the architecture utilizes independent asynchronous stream readers coupled with a dedicated dispatcher. This decoupling isolates the variable network latencies of individual IP cameras from the main processing pipeline. By aligning incoming frames into synchronized paired queues, the system guarantees that the spatial fusion algorithm evaluates identical temporal states, preventing perspective-induced identity mismatches caused by asynchronous cow movements.

2.3.2. Adaptive inference and parallelization

To accommodate the computational load of the behavior-aware detector, the architecture parallelizes YOLOv9s inference across independent threads for each camera view, dynamically utilizing available GPU parallelism. Furthermore, the system implements an adaptive resource management strategy to balance spatial precision with real-time constraints. Rather than executing computationally intensive instance segmentation on every frame, the architecture triggers EdgeSAM mask refinement periodically and only for high-confidence detections. This adaptive allocation ensures that the CowTracker module receives refined spatial contours when occlusion resolution is critical, without causing global pipeline bottlenecks.

2.3.3. Asynchronous spatio-temporal tracking and behavioral analytics

The core tracking logic – encompassing multi-constraint target association, trajectory verification, and Global ID Pool management – is centralized within a dedicated fusion thread. By isolating this spatio-temporal logic from the heavier visual perception tasks, the system maintains strict ID consistency across frames.

Crucially, the ultimate goal of the Bovine-LTID system is long-term behavioral profiling (e.g., time-budget analysis). Writing extensive trajectory coordinates, mask crops, and behavioral statistics to disk synchronously would introduce latency spikes. Therefore, the architecture routes all resultant tracking metadata into decoupled asynchronous data collection queues. Independent analytical threads process these

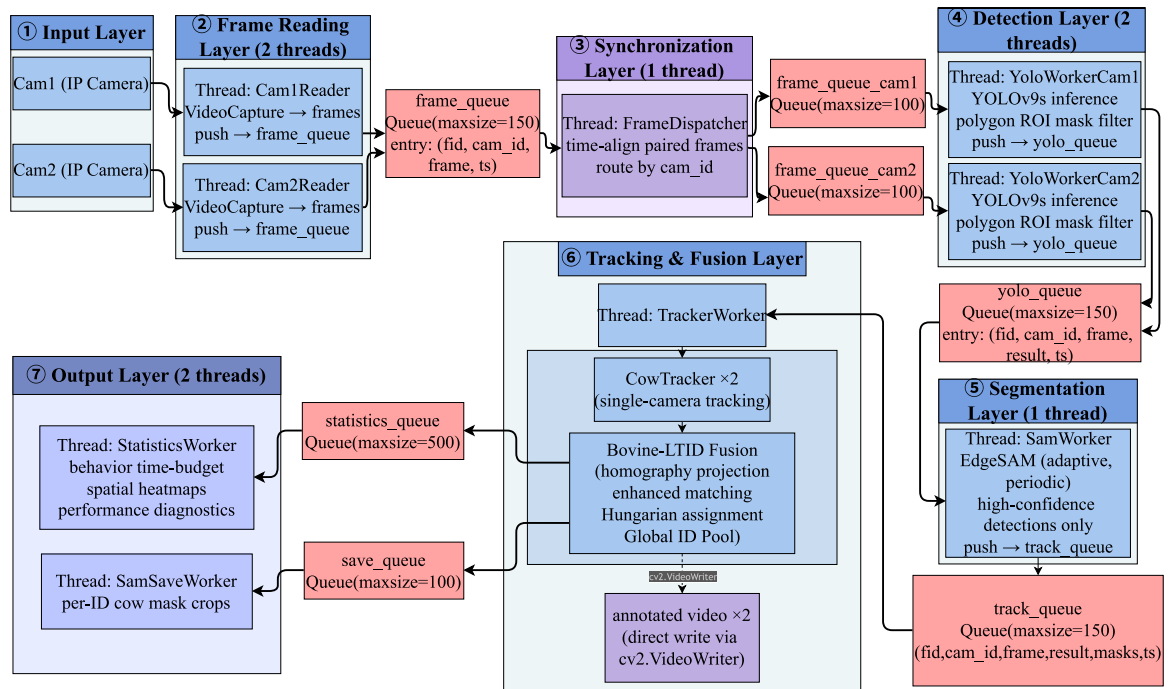


Fig. 4. The proposed concurrent system architecture designed for real-time Bovine-LTID deployment. By isolating stream ingestion, inference, and behavioral data aggregation into asynchronous pipelines, the system maintains temporal synchronization for cross-view fusion while ensuring continuous > 12 FPS throughput on consumer-grade hardware.

Table 4

Per-module latency and peak memory on consumer-grade hardware (NVIDIA RTX 4060 Laptop, 8 GB VRAM).

Pipeline stage	Modules	Avg latency (ms)	Peak latency (ms)	Peak GPU (MB)
<i>Concurrent stages (thread-parallel)</i>				
Frame I/O (2 cams)	FrameReader _{x2}	83.1	588.9	–
YOLOv9 Detection (2 cams)	YOLO Inference + Filter	526.7	614.3	164
EdgeSAM Segmentation	SAM _{x2}	93.1	666.5	176
<i>Serial stages (tracker thread)</i>				
CowTracker Update	2-camera tracking	27.5	223.4	237
Cross-camera Fusion	Matching + Trajectory	1.8	98.4	370
Visualization + Output	Drawing + VideoWrite	52.9	162.7	418
End-to-End				
Effective Throughput	Pipeline bottleneck	12.0 FPS	–	–
Total GPU Memory	Peak allocated	–	–	488
Model Baseline (idle)	Weights only	–	–	97

Measured on a 6000-frame dual-camera sequence (NVIDIA RTX 4060 Laptop, 8 GB). The pipeline uses a 5-stage thread-parallel architecture with per-camera processing queues. YOLOv9 GPU inference processes batches averaging 3.56 frames, completing in 296 ms per batch (≈ 83 ms/frame) on Camera 1; the CPU-side mask filter (230 ms/batch) overlaps with the alternate camera's GPU inference via dual-thread parallelism. Camera 2 exhibits similar timing characteristics (320 ms GPU inference and 251 ms CPU filtering per batch). The GPU inference stage determines the profiled end-to-end throughput at ~ 12.0 FPS. Notably, extensive diagnostic logging during profiling introduces a slight computational overhead; during standard operational deployment, the system maintains approximately 13 FPS, subject to minor dynamic fluctuations based on scene density. Peak GPU memory (488 MB, including the 97 MB model baseline) remains well within consumer GPU capacity. CPU memory peak (USS): 15.8 GB (process working set, including frame queue buffers).

queues in the background to aggregate time-budgets, generate spatial heatmaps, and compile performance diagnostics. This memory-efficient, non-blocking design is what enables Bovine-LTID to continuously monitor herd welfare over extended periods (e.g., multi-hour sessions) without memory exhaustion or temporal fragmentation.

To empirically validate the feasibility of deployment on consumer-grade hardware, we profiled the per-module latency and peak memory utilization of the system on an Intel i9-14900HX CPU and an NVIDIA RTX 4060 Laptop GPU. As detailed in Table 4, the concurrent architecture successfully isolates the heavier visual perception workloads from the serial tracking logic. The total GPU memory footprint during active processing peaks at only 488 MB (391 MB incremental over the 97 MB model baseline). Governed by the YOLO detection bottleneck, the system maintains a profiled end-to-end throughput of 12.0 FPS.

Because the intensive diagnostic logging required for module-level profiling introduces minor computational overhead, standard operational deployment yields a practical processing speed of approximately 13 FPS (subject to minor scene-dependent fluctuations). This confirms that robust, multi-camera individual cattle monitoring can be deployed efficiently on standard edge devices, providing a scalable solution for precision livestock farming without requiring enterprise-grade computing clusters.

3. Methodology: Spatio-temporal tracking and fusion

3.1. System initialization: Homography calibration and ROI masking

Prior to tracking operations, precise camera coordinate alignment and spatial constraints are established through a systematic calibration

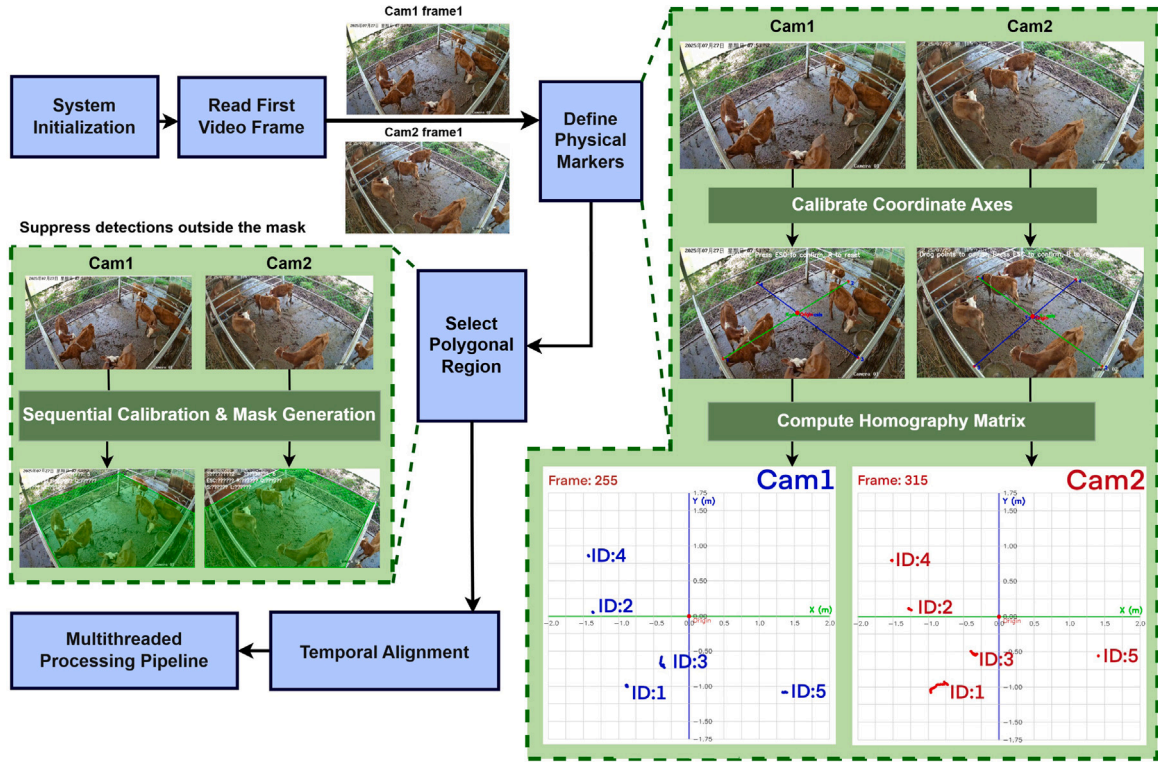


Fig. 5. Homography-based coordinate transformation from pixel to world coordinates with ROI masking. Pre-marked physical center points ensure consistent, high-precision calibration across experimental sessions. The visualization demonstrates how image coordinates from two camera perspectives are unified into a common world coordinate system, enabling accurate multi-view cattle tracking and behavior analysis.

protocol designed for operational robustness in agricultural environments. To ensure reproducible and high-precision calibration, four central reference points of the monitoring area were precisely marked in the physical barn environment using measuring tape and permanent markers, with known world coordinates forming a rectangular reference frame that corresponds to the typical cattle activity zone. This standardized coordinate system provides a consistent spatial reference across all experimental sessions and camera viewpoints (see Fig. 5).

During system initialization, the corresponding image coordinates of these **pre-defined physical markers** are specified for each camera view using fixed coordinate arrays. These carefully measured point correspondences form the foundation for computing the homography matrix H_k for each camera k via Direct Linear Transformation (DLT) (Hartley and Zisserman, 2004), which establishes the projective transformation between image coordinates (u, v) and world coordinates (X, Y) :

$$\begin{bmatrix} X' \\ Y' \\ W \end{bmatrix} = H_k \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}, \quad (X, Y) = \left(\frac{X'}{W}, \frac{Y'}{W} \right). \quad (2)$$

This marker-based calibration methodology offers several critical advantages for livestock monitoring applications. First, it eliminates inter-operator variability that could introduce systematic errors in multi-session experiments. Second, it ensures sub-meter alignment accuracy essential for reliable cross-camera identity matching. Third, the physical marker approach provides inherent resilience to camera repositioning and environmental changes, as the ground reference frame remains constant.

While the rigidly ceiling-mounted cameras and hardcoded coordinates prevent algorithmic calibration drift, commercial barns are subject to physical disturbances. To ensure long-term robustness, the physical anchor points serve as a permanent ground truth. Any physical shift in the camera perspective can be detected by periodically verifying the actual positions of the physical floor markers against their

hardcoded pixel coordinates. A practical deployment and maintenance checklist is provided in Appendix A.1.

To restrict detection processing to relevant cattle activity zones, the system implements a polygon-based region of interest (ROI) masking mechanism. The ROI boundaries are strategically defined to correspond with physical structural elements and functional areas within the barn environment. The core purpose of this spatial filtering is to prevent interference from cattle in adjacent pens, ensuring the system tracks and analyzes only the designated herd. The ROI configuration system features persistent storage in JSON format, supporting dynamic reconfiguration during operation to facilitate adaptive monitoring.

3.2. Single-camera tracking: CowTracker

Cattle in each camera view are tracked using CowTracker, an enhanced ByteTrack-based algorithm designed specifically for livestock monitoring. Unlike generic multi-object trackers, CowTracker explicitly addresses challenges unique to barn environments, including frequent occlusions, visually similar individuals, irregular motion dynamics, and strong lighting variations.

Built upon the ByteTrack framework (Zhang et al., 2022) (Fig. 6), CowTracker integrates four major algorithmic enhancements that improve matching robustness, trajectory continuity, identity stability, and global ID consistency. Note that all specific hyperparameter values utilized in our experiments are comprehensively detailed in Appendix D.

1. **Multi-constraint matching with trajectory-aware scoring (E1):** Rather than relying solely on IoU, CowTracker jointly considers spatial overlap, centroid proximity, and a trajectory-derived robustness score. The association score is:

$$S_{\text{match}} = \begin{cases} \lambda_1 \text{IoU}(b_i, b_j) + \lambda_2 \left(1 - \frac{D_{\text{centroid}}}{D_{\text{dyn}}} \right) + \lambda_3 S_{\text{robust}}, & \text{if robust cues available,} \\ \gamma_1 \text{IoU}(b_i, b_j) + \gamma_2 \left(1 - \frac{D_{\text{centroid}}}{D_{\text{dyn}}} \right), & \text{otherwise.} \end{cases}$$

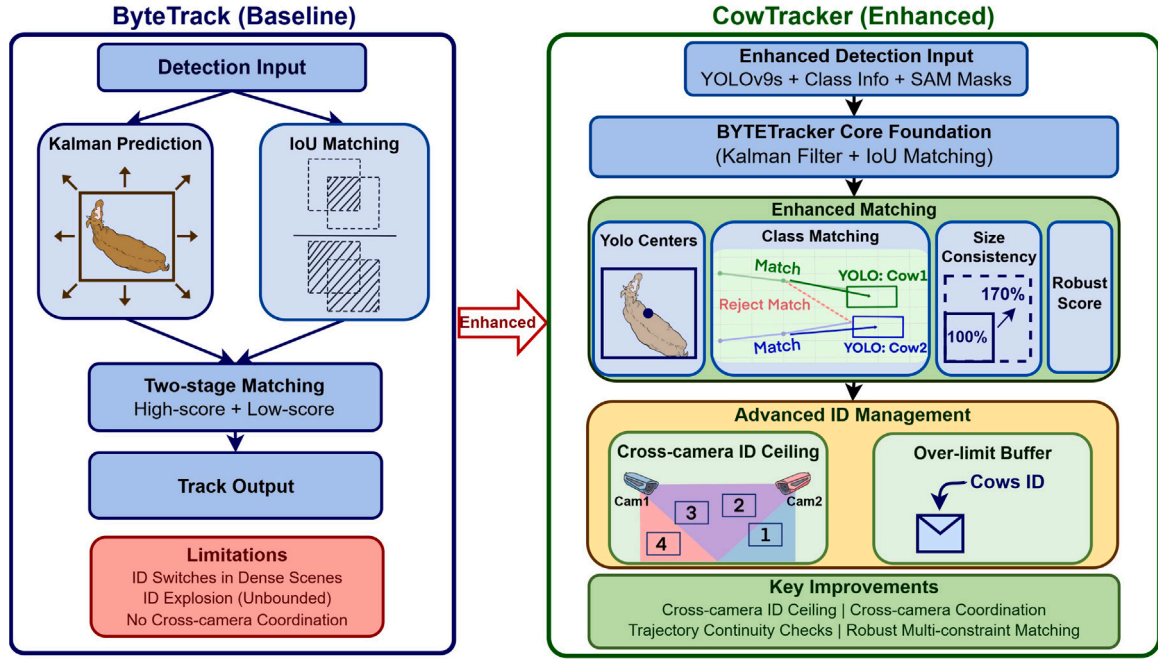


Fig. 6. High-level conceptual comparison between the baseline ByteTrack and the proposed CowTracker. CowTracker overcomes the limitations of generic trackers by introducing multi-constraint matching, ID-stability protection, adaptive trajectory verification, and cross-camera ID ceiling management tailored for dense barn environments.

(3)

where λ_k and γ_k are normalized weighting coefficients. The dynamic distance threshold D_{dyn} adapts to target scale and video frame rate. The robustness term $S_{\text{robust}} = w_p S_{\text{pos}} + w_s S_{\text{size}} + w_d S_{\text{dir}}$ combines position prediction error (S_{pos}), area stability (S_{size}), and trajectory direction cosine similarity (S_{dir}) parameterized by corresponding weights.

2. **Adaptive trajectory continuity verification (E2):** To prevent implausible identity jumps during occlusions, CowTracker applies a two-gate check before every ID assignment:

$$\begin{cases} D_{\text{jump}} \leq D_{\text{jump}}^{\text{th}}(\Delta t), \\ V_{\text{current}} \leq \max(V_{\text{max}}, m \cdot \bar{V}_{\text{history}}). \end{cases} \quad (4)$$

The jump threshold $D_{\text{jump}}^{\text{th}}$ is dynamically scaled based on the detection frame gap Δt , accommodating larger spatial displacements during prolonged occlusions. The velocity ceiling prevents biologically implausible acceleration by bounding the instantaneous velocity V_{current} against an absolute maximum V_{max} and a historical track average $\bar{V}_{\text{history}}$ scaled by a multiplier m .

3. **ID-stability protection with explicit rejection rule (E3):** To suppress oscillatory identity switches in crowded scenes, CowTracker explicitly rejects any candidate identity assignment if the target identity has undergone multiple distinct switches (N_{switch}) within a recent temporal window (W_{frames}). Candidates passing this stability gate are further evaluated on class-label consistency and size stability.
4. **Cross-camera ID ceiling with over-limit buffering (E4):** A dynamic ID ceiling $N_{\text{ceiling}} = \max(N_{\text{current}}, N_{\text{cross}})$ is maintained, where N_{current} is the active local tracks count and N_{cross} is the shared upper bound from the overlapping view. Any assigned ID exceeding N_{ceiling} is placed into a short-term over-limit buffer (B_{over}) for F_{buffer} frames. If the ceiling subsequently rises, buffered identities are restored; otherwise, they expire.

The overall workflow is formalized in **Algorithm 1** (detailed in Appendix F) and visualized in Fig. 7.

The four enhancement blocks are complemented by the periodic EdgeSAM mask refinement introduced in the detection pipeline. Within CowTracker, the mask m_i provides **SAM ID mapping** for post-occlusion re-association and supports **temporal mask-consistency weighting** in the cross-view behavior inference stage. Neither function is a structural dependency of the core spatio-temporal matching logic, which operates predominantly on bounding-box geometry and motion cues.

In the design of CowTracker, we deliberately favored explicit, manual spatial heuristics over modern adaptive or learned spatial models. For automated livestock welfare monitoring, diagnostic interpretability is crucial; explicit kinematic bounds prevent the unpredictable identity jumps sometimes exhibited by “black-box” adaptive models during severe mutual occlusions. Furthermore, the rigid physical constraints of our monitoring setup render highly parameterized adaptive spatial learning computationally redundant for edge deployment. As demonstrated by the empirical sensitivity analysis (detailed in Section 4.2.5), these selected geometric thresholds exhibit strong stability and are robust to minor parameter shifts.

3.3. Multi-camera identity fusion and cross-view behavior inference

3.3.1. Cross-camera identity matching via enhanced world coordinate fusion

Track positions from each camera are transformed into a unified ground-plane coordinate system using pre-calibrated homography matrices H_1 and H_2 :

$$C_i^k = H_k \cdot (u_i^k, v_i^k, 1)^T, \quad k \in \{1, 2\} \quad (5)$$

where C_i^k represents the world coordinates of target i from camera k .

3.3.2. World-distance Hungarian matching

For initial cross-camera data association, the system constructs a spatial cost matrix based on pairwise Euclidean distances between world-projected detection centers. The Hungarian algorithm (Kuhn, 1955) solves the optimal assignment:

$$\min \sum_{i=1}^{N_1} \sum_{j=1}^{N_2} \|C_i^1 - C_j^2\| \cdot x_{ij}, \quad \text{subject to } \sum_i x_{ij} = 1, \sum_j x_{ij} = 1 \quad (6)$$

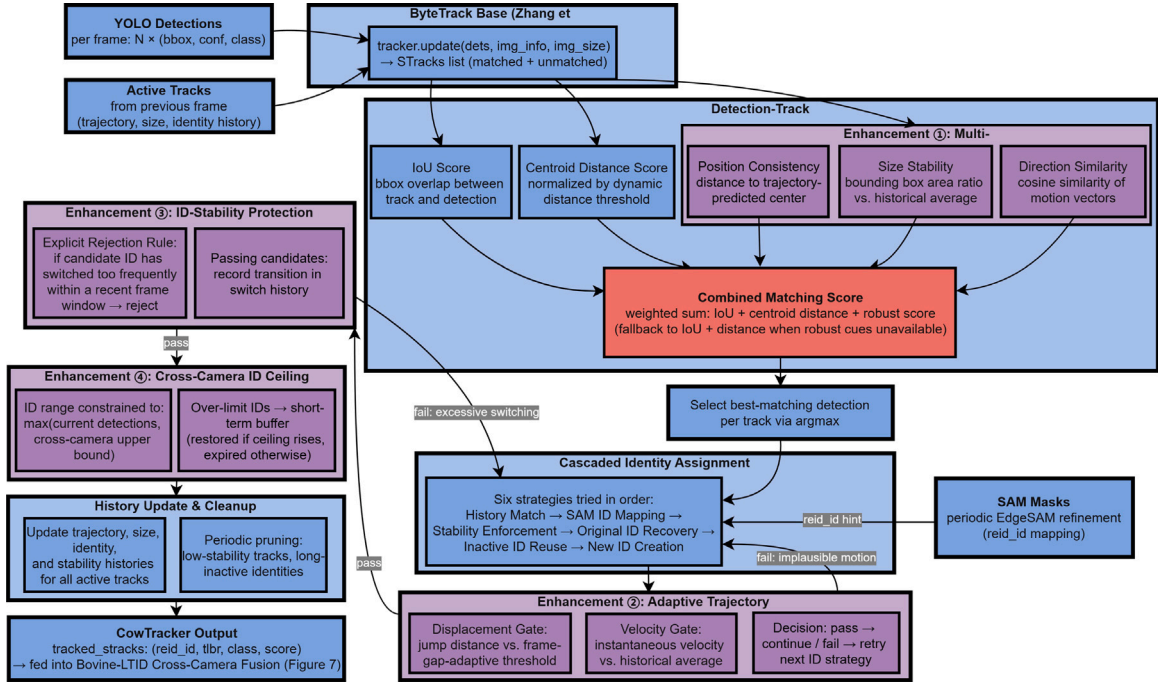


Fig. 7. Detailed algorithmic flow diagram of CowTracker. The four core enhancement blocks (E1–E4) are strategically inserted into the baseline ByteTrack association pipeline.

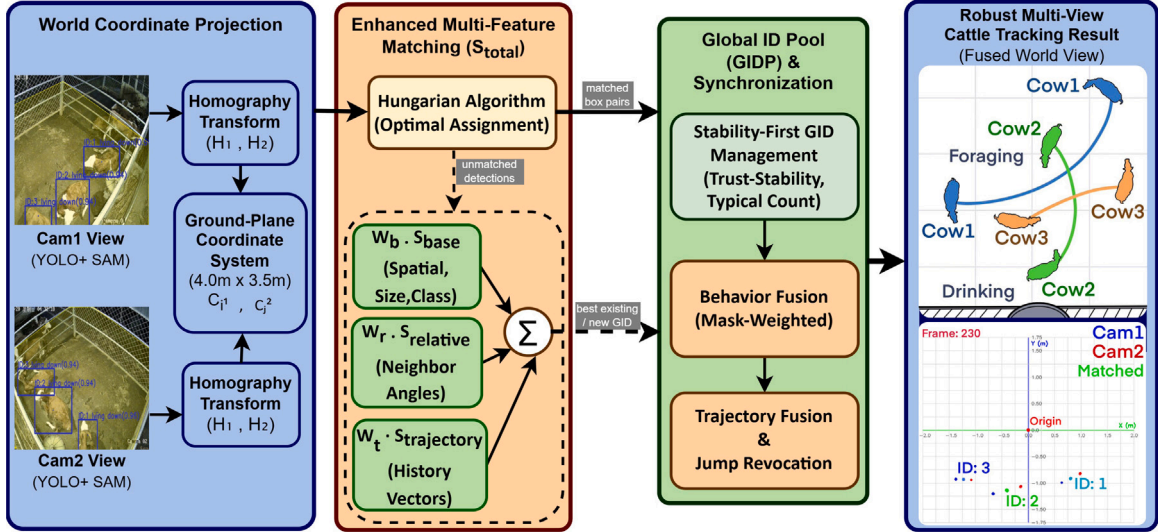


Fig. 8. Cross-camera identity fusion pipeline: world-distance Hungarian matching generates candidate associations; enhanced multi-feature matching re-associates unmatched tracks; GIDP resolves identities.

Candidate pairs whose world distance exceeds a maximum spatial threshold $d_{\max}$ are assigned infinite cost. Pairs satisfying a minimum spatial similarity τ_{sim} are accepted and forwarded to the Global ID Pool (GIDP).

3.3.3. Enhanced multi-feature re-association for unmatched tracks

Detection boxes remaining unassigned are processed through an enhanced multi-feature matching algorithm, integrating three complementary similarity measures (see Fig. 8):

$$S_{\text{total}} = w_b \cdot S_{\text{base}} + w_r \cdot S_{\text{relative}} + w_t \cdot S_{\text{trajectory}} \quad (7)$$

where w_b, w_r, w_t are weighting coefficients governing the relative importance of each feature.

Base Similarity (S_{base}) combines spatial proximity, size consistency, and class agreement:

$$S_{\text{base}} = w_d \cdot \max\left(0, 1 - \frac{\|C_i^1 - C_j^2\|}{d_{\max}}\right) + w_s \cdot \frac{\min(A_i, A_j)}{\max(A_i, A_j)} + w_c \cdot \mathbb{I}(c_i = c_j) \quad (8)$$

Relative Position Consistency (S_{relative}) validates spatial relationships with neighboring targets:

$$S_{\text{relative}} = \frac{1}{N} \sum_{k=1}^N \max(0, \cos(\theta_k^1 - \theta_k^2)) \quad (9)$$

To avoid implausible vertical mismatches common in overhead views, we enforce a dynamic Y-coordinate difference threshold $\Delta Y_{\max}$.

Trajectory Consistency ($S_{\text{trajectory}}$) evaluates motion pattern alignment using recent historical trajectories (window size L):

$$S_{\text{trajectory}} = \frac{1}{M} \sum_{m=1}^M \max(0, \cos(\phi_m^1 - \phi_m^2)) \quad (10)$$

If the best match score exceeds an acceptance threshold τ_{reassoc} , the unmatched track inherits the corresponding global identity.

3.3.4. Global ID management and synchronization

A centralized Global ID Pool (GIDP) maintains consistent identity across camera views. To enhance long-term identity stability, we maintain a short-term history of local-to-global ID bindings. Specifically, when one camera maintains stable tracks (e.g., F_{stable} consecutive frames) while the other undergoes transient occlusions, the system prioritizes the GID assignments from the stable viewpoint. A dynamic ID compaction mechanism periodically prunes obsolete high-numbered IDs.

3.3.5. Cross-view behavior inference with mask-consistency weighting

Behavior predictions from individual cameras are fused using a confidence-weighted approach enhanced by segmentation mask consistency:

$$\text{conf}_{\text{enhanced}} = (1 - w_m) \cdot \text{conf}_{\text{YOLO}} + w_m \cdot \text{IoU}_{\text{mask}} \quad (11)$$

where w_m governs the mask contribution.

3.3.6. Trajectory fusion and multi-resolution analysis

World coordinates from matched targets are fused via weighted averaging. The system maintains dual trajectory representations: Dense recording for comprehensive heatmap analysis (spatial grid resolution) and Sparse visualization (filtered by a minimum displacement threshold) to ensure clean trajectory plots. Any trajectory update exceeding a maximum jump filter distance is rejected to prevent matching artifacts.

3.3.7. Real-time performance and memory optimization

To maintain real-time performance, the fusion system implements synchronized frame processing, adaptive data structures (e.g., bounded deques with max length L_{queue}), memory-efficient operations, and selective SAM processing for high-confidence detections.

4. Results and analysis

The evaluation of Bovine-LTID is conducted in three phases: (1) frame-level behavior-aware detection accuracy on a held-out annotated test set; (2) single-camera tracking quality benchmarked against five state-of-the-art trackers; and (3) cross-camera identity fusion evaluated on a dual-camera configuration. For tracking evaluation, we employ a dual-metric framework on the same 6000-frame sequence. *Ground-truth* (GT) metrics are computed directly against manually annotated bounding-box identities via Hungarian matching. *Proxy metrics*, derived from spatio-temporal self-consistency criteria without external labels (defined in Appendix E), are computed on the identical data. Reporting both on the same sequence allows direct assessment of whether the proxy framework, which is applicable to arbitrarily long unlabeled recordings, captures the same trends as the gold-standard GT evaluation.

4.1. YOLO-based behavior-aware cattle detection

We evaluated the proposed Bovine-LTID against 17 YOLO variants. To ensure a fair comparison, all models were fine-tuned using an identical protocol on a single NVIDIA H200 GPU (operating within an 8× H200 cluster for parallel processing). As summarized in Table 5, Bovine-LTID achieves a competitive mAP50 of 0.944 and an mAP50-95 of 0.885. The results indicate a strong balance between Precision (96.1%) and Recall (94.0%), while maintaining a practical inference speed of 57.9 FPS.

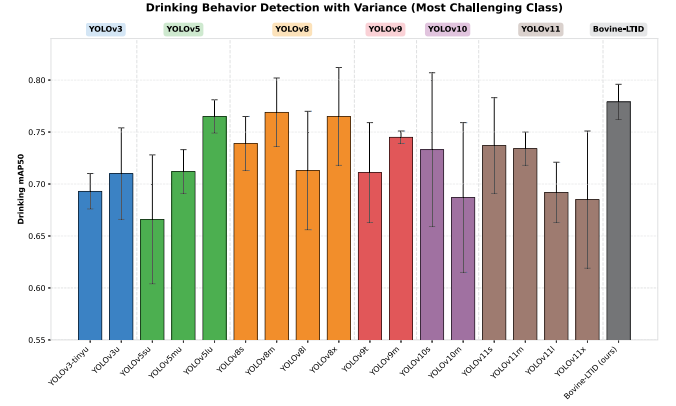


Fig. 9. Granular comparison of mAP50 for Drinking behavior. Bovine-LTID demonstrates improved robustness in handling head-down occlusions compared to baseline YOLO models.

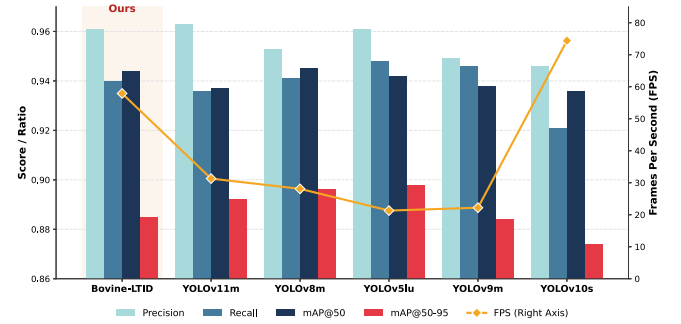


Fig. 10. Multi-metric comparison of the top 6 models. Bovine-LTID shows competitive overall detection quality with a strong Precision–Recall balance.

4.1.1. Targeted robustness in challenging behavioral contexts

A per-class analysis indicates that while most models perform consistently well on distinct stationary postures (such as “Stand” and “Lying”), notable performance disparities emerge during occlusion-prone behaviors.

Fig. 9 focuses on the Drinking class, which proved to be the most challenging category across evaluations. While several variants, such as YOLOv3-tinyu (0.693) and YOLOv8l (0.713), exhibit lower accuracy in this context, Bovine-LTID yields an mAP50 of 0.779. This improvement helps mitigate the visual confusion between cattle in drinking and standing postures.

4.1.2. Comprehensive multi-metric performance assessment

Fig. 10 further contextualizes these results among the top-performing models. Bovine-LTID maintains a consistent overall detection quality, offering an effective trade-off between mAP metrics and the Precision–Recall balance. This reliable localization serves as a crucial foundational step for the subsequent spatio-temporal tracking pipeline.

4.2. Single-camera tracking evaluation

We evaluate single-camera tracking on a 6000-frame sequence with manually annotated bounding-box identities. Six trackers are compared using identical YOLOv9 detections: ByteTrack (Zhang et al., 2022), DeepSORT (Wojke et al., 2017), StrongSORT (Du et al., 2023), OC-SORT (Cao et al., 2023), Deep OC-SORT (Maggiolino et al., 2023), and the proposed CowTracker. For fairness, CowTracker’s ID recovery module is disabled, aligning its configuration with baselines that lack post-occlusion re-association mechanisms.

Table 5

Comprehensive quantitative comparison of 17 YOLO variants and the proposed Bovine-LTID on the held-out test set.

Model	Per-Class mAP50 (↑)					Overall metrics				
	Stand	Lying	Foraging	Drinking	Unclear	mAP50	mAP50-95	P (%)	R (%)	FPS
YOLOv3-tinyu	0.979	0.995	0.981	0.693 ± 0.017	0.991	0.928 ± 0.002	0.825 ± 0.008	94.2 ± 0.4	93.8 ± 1.3	93.6 ± 1.0
YOLOv3u	0.981	0.995	0.978	0.710 ± 0.044	0.994	0.931 ± 0.009	0.891 ± 0.007	95.2 ± 1.2	92.1 ± 1.2	11.3 ± 0.0
YOLOv5su	0.979	0.992	0.984	0.666 ± 0.062	0.993	0.923 ± 0.010	0.867 ± 0.007	94.7 ± 1.3	94.3 ± 0.8	84.7 ± 0.2
YOLOv5mu	0.980	0.995	0.987	0.712 ± 0.021	0.993	0.933 ± 0.004	0.884 ± 0.009	94.1 ± 1.1	93.5 ± 1.3	34.1 ± 0.1
YOLOv5lu	0.981	0.995	0.974	0.765 ± 0.016	0.993	0.942 ± 0.003	0.898 ± 0.004	96.1 ± 1.0	94.8 ± 0.2	21.3 ± 0.0
YOLOv8s	0.980	0.995	0.977	0.739 ± 0.026	0.994	0.937 ± 0.005	0.879 ± 0.003	94.1 ± 2.9	94.0 ± 1.2	75.6 ± 0.3
YOLOv8m	0.984	0.995	0.983	0.769 ± 0.033	0.992	0.945 ± 0.008	0.896 ± 0.009	95.3 ± 1.6	94.1 ± 0.7	28.1 ± 0.0
YOLOv8l	0.981	0.995	0.971	0.713 ± 0.057	0.994	0.931 ± 0.013	0.882 ± 0.019	94.8 ± 1.5	92.9 ± 1.7	18.1 ± 0.0
YOLOv8x	0.983	0.995	0.984	0.765 ± 0.047	0.993	0.944 ± 0.010	0.893 ± 0.009	95.2 ± 0.5	94.1 ± 1.1	8.5 ± 0.0
YOLOv9t	0.981	0.995	0.981	0.711 ± 0.048	0.993	0.932 ± 0.009	0.865 ± 0.010	93.1 ± 1.5	94.6 ± 0.9	73.8 ± 0.9
YOLOv9m	0.977	0.995	0.980	0.745 ± 0.006	0.993	0.938 ± 0.001	0.884 ± 0.004	94.9 ± 0.1	94.6 ± 0.1	22.2 ± 0.0
YOLOv10s	0.978	0.995	0.981	0.733 ± 0.074	0.993	0.936 ± 0.014	0.874 ± 0.017	94.6 ± 1.7	92.1 ± 1.6	74.4 ± 0.6
YOLOv10m	0.983	0.995	0.977	0.687 ± 0.072	0.993	0.927 ± 0.015	0.877 ± 0.019	94.0 ± 3.8	91.5 ± 3.2	31.3 ± 0.2
YOLOv11s	0.981	0.995	0.990	0.737 ± 0.046	0.993	0.939 ± 0.008	0.880 ± 0.015	94.1 ± 1.6	93.1 ± 1.9	74.0 ± 0.2
YOLOv11m	0.983	0.995	0.981	0.734 ± 0.016	0.993	0.937 ± 0.003	0.892 ± 0.000	96.3 ± 0.1	93.6 ± 0.8	31.3 ± 0.0
YOLOv11l	0.983	0.995	0.981	0.692 ± 0.029	0.991	0.928 ± 0.005	0.887 ± 0.003	94.4 ± 1.4	93.7 ± 0.7	25.1 ± 0.1
YOLOv11x	0.980	0.995	0.985	0.685 ± 0.066	0.991	0.927 ± 0.013	0.870 ± 0.018	94.1 ± 1.5	92.6 ± 1.9	12.0 ± 0.1
Bovine-LTID (ours)	0.979	0.995	0.976	0.779 ± 0.017	0.992	0.944 ± 0.003	0.885 ± 0.008	96.1 ± 0.3	94.0 ± 0.7	57.9 ± 0.3

Note: All models were fine-tuned on an NVIDIA H200 GPU. Accuracy metrics (mAP, P, R) are evaluated over 3 random seeds (0, 42, 123) with ± denoting standard deviation. Per-class mAP50 for Stand, Lying, Foraging, and Unclear report mean only ($\sigma \leq 0.005$ across all models). To reflect real-world edge deployment feasibility, raw inference speed (FPS) was explicitly measured in FP16 mode at 1080p resolution on a consumer-grade NVIDIA RTX 4060 Laptop GPU, averaged over 5 rounds.

Table 6

Single-camera tracking quality against manually annotated ground truth (6000-frame subset). Values are presented as Camera 1/Camera 2.

Method	F1 (GT) ↑	IDF1 (GT) ↑	ID Sw/1k ↓	Wtd. LLTC (%) ↑	Long-Trk (%) ↑
ByteTrack	0.9910/.9870	0.8011/.7726	10.3/11.2	55.3/47.0	97.1/96.4
DeepSORT	0.9726/.9665	0.7428/.6609	10.3/ 9.5	44.4/39.3	92.9/90.4
StrongSORT	0.9932/.9883	0.7009/.7107	21.2/12.0	38.0/40.0	97.1/93.8
OC-SORT	0.9869/.9840	0.5814/.6774	11.8/ 9.5	27.0/35.9	91.8/95.6
Deep OC-SORT	0.9867/.9841	0.6818/.6705	13.0/10.8	40.4/32.8	91.3/94.2
CowTracker (ours)	0.9901/.9867	0.8001/.7698	17.5/25.5	55.1/47.3	97.9/97.2

CowTracker with ID recovery disabled. GT = ground truth.

4.2.1. Ground-truth evaluation

All metrics are computed by Hungarian-matching system detections to manual annotations at IoU ≥ 0.5 , without requiring identity-number agreement. **F1** is the per-frame detection-level harmonic mean of Precision and Recall. **IDF1** associates each annotated cow with its most frequent system track ID across frames and computes the F1 of these associations. **ID switch rate** (per 1000 frames) counts transitions between different system track IDs for each GT cow. **Weighted LLTC** is the frame-weighted mean temporal coverage of system tracks: $\text{LLTC} = \sum_i w_i c_i$, where $c_i = n_i/T$ and $w_i = n_i / \sum_j n_j$. **Long-track coverage** is the fraction of tracking time from tracks lasting ≥ 30 s.

Results are summarized in Table 6. CowTracker achieves competitive weighted LLTC (55.1% / 47.3%) and Long-Track Coverage (97.9% / 97.2%) among the evaluated methods. Its F1 and IDF1 scores are comparable to ByteTrack (a strong motion-only baseline), indicating that livestock-specific spatio-temporal heuristics support identity consistency without relying on appearance feature extraction.

4.2.2. Proxy evaluation on the unannotated sequence

For extensive video sequences where exhaustive manual identity annotation is impractical, we introduce a proxy evaluation framework (detailed in Appendix E) that assesses tracking quality through spatio-temporal self-consistency rather than external ground-truth labels. Physical trajectories are reconstructed by greedily chaining detections via bounding-box IoU across consecutive frames. Proxy metrics are then derived from the coherence of the system-assigned local IDs within these continuous physical tracks.

The proxy results (Table 7) mirror the trends observed in the ground-truth validation. CowTracker and ByteTrack maintain robust identity coherence, while ReID-based methods exhibit varied performance, likely influenced by the visual homogeneity of the cattle.

4.2.3. Algorithmic trade-offs and tracklet safety

Both Tables 6 and 7 reveal that when the identity recovery module is deactivated for single-camera benchmarking, CowTracker exhibits a higher identity switch rate (17.5/25.5 per 1k under GT; 27.7/29.7 per 1k under Proxy) and lower Long-Track Coverage compared to ByteTrack. These single-view metrics reflect a conservative association philosophy embedded within the CowTracker gating layer. To suppress erroneous identity assignments during dense mutual occlusions or transient detection drift, CowTracker enforces strict multi-constraint verification, including spatial displacement jump limits and scale consistency. When these kinematic criteria are violated, the system terminates the tracking state and generates a new local identity rather than risking a false-positive connection between distinct animals. Consequently, while looser association criteria can yield longer continuous segments within a single isolated viewpoint, they exhibit a vulnerability to undetected cross-target identity contamination. CowTracker's intentional localized fragmentation preserves the purity of the tracklet assignments by ensuring a low false-positive correlation rate. The structural safety of these fragmented tracklets is quantitatively validated during the subsequent multi-view integration phase. As demonstrated by the cross-camera fusion results, when these local segments are processed by the centralized Global ID Pool (GIDP) and its over-limit buffering, the aggregate raw global switch rate is measurably reduced. This trend contrasts with the baseline projective fusion, where unverified tracking data lead to global trajectory deterioration. Ultimately, these single-camera local tracklets establish a reliable foundation for continuous herd-level behavioral analytics across overlapping fields of view.

4.2.4. Relationship between GT and proxy metrics

It is critical to distinguish the distinct constructs evaluated by the two independent verification frameworks. Ground-Truth (GT) metrics

Table 7

Proxy single-camera evaluation on the full sequence. These metrics are derived from self-consistency criteria (measuring physical track coherence) and serve as relative performance indicators rather than absolute accuracy against ground truth. Values are presented as Camera 1/Camera 2.

Method	Proxy IDF1 (proxy) $\uparrow$	Wtd. LLTC (%) (proxy) $\uparrow$	Occ. Recov. (%) (proxy) $\uparrow$	ID Sw/1k (proxy) $\downarrow$	Long-Trk (%) (proxy) $\uparrow$
ByteTrack	0.97/0.99	22.5/17.9	81.0/72.9	4.9/ 5.5	64.4/65.3
DeepSORT	0.98/0.97	12.2/13.9	84.2/83.1	3.8/ 2.7	64.0/63.1
StrongSORT	0.89/0.96	16.3/18.2	83.5/83.6	11.0/10.3	57.9/57.4
OC-SORT	0.81/0.88	9.9/11.2	85.3/85.8	10.3/ 9.0	52.8/50.7
Deep OC-SORT	0.83/0.89	10.5/14.3	86.8/85.5	10.5/ 9.2	54.6/52.4
CowTracker (ours)	0.97/0.96	22.5/17.9	78.6/78.0	27.7/29.7	42.0/45.7

Proxy metrics defined in Appendix E. CowTracker with ID recovery disabled.

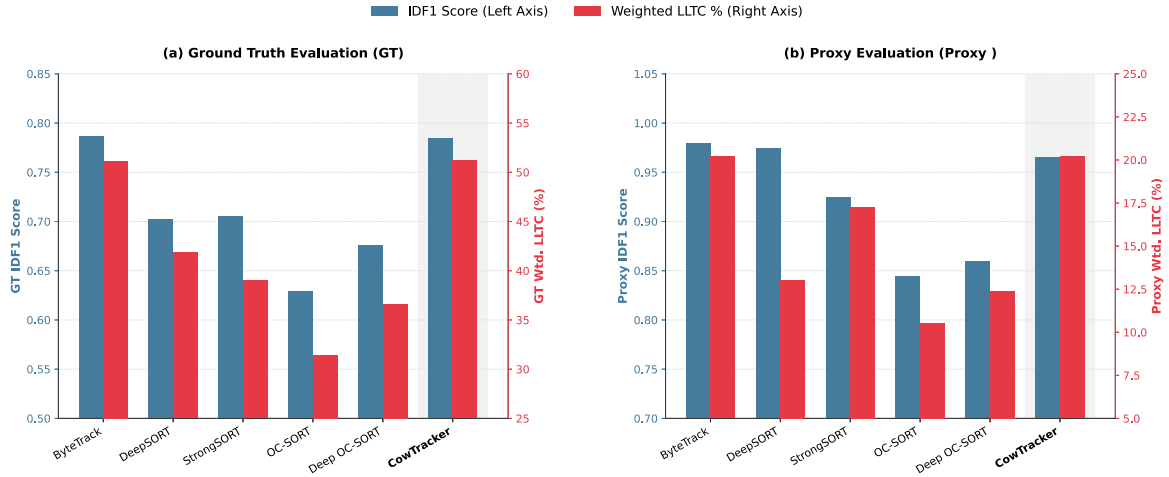


Fig. 11. Side-by-side performance profiling across evaluation frameworks. (a) Ground Truth evaluation measures absolute accuracy. (b) Proxy evaluation measures spatio-temporal self-coherence. Despite differing absolute values due to fundamentally different constructs, the relative method rankings exhibit consistent alignment.

quantify absolute assignment accuracy against human-annotated animal identities, whereas Proxy metrics assess the internal self-coherence of system-assigned local identifiers within physically continuous spatial trajectories. Consequently, direct numerical comparisons between the absolute values of the two frameworks are structurally invalid. For instance, a tracking model could yield a high Proxy IDF1 by maintaining stable tracklet states internally, even if those tracklets are associated with incorrect global identifiers relative to the human-annotated baseline.

Despite these structural differences in measurement definitions, the relative performance topography among the evaluated tracking methods exhibits consistency across both evaluation paradigms (Fig. 11). The relative ranking profile – where motion-based tracking methods maintain favorable positions in both tracking coherence (IDF1) and temporal coverage (LLTC) – is preserved between the two subsets. This qualitative alignment indicates that while proxy metrics do not substitute for manual ground-truth evaluations in absolute accuracy benchmarking, they provide a reliable indicator for relative method ranking, parameter optimization, and tracking stability analysis on large-scale, unannotated agricultural surveillance streams.

4.2.5. Ablation study of CowTracker components

To understand the mechanical contribution of the proposed CowTracker enhancements (E1–E4), we conducted a component ablation study, summarized in Table 8. The analysis records internal activation counters alongside both local and global tracking metrics to demonstrate the relationship between functional filtering and trajectory stability.

The empirical data indicate an architectural synergy between local tracking conservatism and global fusion. At the single-view level, the local tracking metrics remain intentionally invariant across ablations, preserving clean tracklets with approximately 111 and 87 switches per

1000 frames for Camera 1 and Camera 2, respectively. Concurrently, the internal activation counters verify that each component actively intercepts unstable geometric states during these localized tracking phases.

When integrated with the global fusion pipeline, this local filtering translates into a reduction in the global semantic switch rate (down to 7.70 per 1000 frames), producing a consistent statistical effect size (Cohen's $d = 1.69$). By comparison, the H+NN baseline, which lacks these verified local inputs, experiences trajectory deterioration resulting in 53.90 raw global switches ($d = -0.79$). These variations clarify that the tracking stability of the system emerges structurally from the integrated ReID-free spatio-temporal logic, rather than depending on a single specific heuristic.

4.3. Cross-camera identity fusion evaluation

We evaluate the complete Bovine-LTID pipeline (comprising CowTracker with identity recovery and cross-camera fusion) on the dual-camera surveillance sequence. To isolate and quantify the performance variations contributed by our centralized synchronization and spatio-temporal logic, we implement a standard baseline framework as requested during peer review (**H+NN Baseline**). This reference pipeline utilizes standard single-view ByteTrack tracking, maps coordinates exclusively via ground-plane homography projection, and resolves cross-view target association through basic nearest-neighbor (NN) Euclidean distance assignment solved by the Hungarian algorithm, bypassing the centralized Global ID Pool (GIDP) stability constraints, multi-feature alignment scoring, and tracking jump revocations.

To facilitate methodological transparency, evaluations are executed under two parallel paradigms on identical video segments: a manual Ground-Truth (GT) validation framework conducted on a continuous 6000-frame human-annotated validation subset, and a Proxy self-consistency framework extended over the complete 10-minute unannotated recording.

Table 8

Ablation study dissecting CowTracker components. Functional validation confirms component activation (E1–E4 counters), demonstrating that while single-view metrics maintain local invariance (retaining localized switches to preserve tracklet purity), these filters collectively yield measurable global cross-camera tracking stability relative to the baseline.

Configuration	Functional verification (activation counters)				local invariance Local Sw/1k (Cam1/Cam2)	Global fusion gain		
	E1 (Robust)	E2 (Block)	E4 (Buffer)	New ID		Global RawSw/1k	Global SemSw/1k	Cohen's <i>d</i>
Full Bovine-LTID (E1+E2+E3+E4)	34,790	1554	796	240	111.90/87.00	9.80	7.70	1.684
– E2 (no TrajFilter)	34,790	0	767	174	111.40/86.50	10.30	8.20	1.677
– E3 (no Stability)	34,790	1553	819	239	111.50/87.00	9.50	7.30	1.690
– E4 (no OverLimit)	34,790	1567	2156	258	111.00/87.50	9.50	7.30	1.687
– E1 (no EnhancedMatch)	0	1503	747	231	110.90/87.50	8.20	6.20	1.691
H+NN Baseline	0	0	134	1003	54.80/36.50	53.90	51.90	–0.791

Table 9

Cross-camera tracking consistency and trajectory stabilization performance validated against manual ground truth (6000-frame validation subset). Switch rates are normalized per 1000 frames to ensure mathematical alignment across localized single-view inputs and globally fused system outputs.

Evaluation phase & configuration	X-cam Consist. (%) ↑	Raw Sw/1k ↓	Semantic Sw/1k ↓	LLTC/GLTC (%) ↑	MT (%) ↑	Cohen's <i>d</i>
<i>Local Single-View Tracking Logs (Per-Camera Tracklets)</i>						
H+NN Baseline – Camera 1 (Local)	–	54.80	28.3	19.90 (LLTC)	–	–
H+NN Baseline – Camera 2 (Local)	–	36.50	10.3	17.80 (LLTC)	–	–
Bovine-LTID Framework – Camera 1 (Local)	–	100.70	81.8	19.50 (LLTC)	–	–
Bovine-LTID Framework – Camera 2 (Local)	–	73.80	60.3	17.40 (LLTC)	–	–
<i>Central Multi-Camera Fusion Phase (Global Trajectories)</i>						
H+NN Baseline (Projective Vision)	99.19	53.90	51.90	83.12 (GLTC)	50.00	–0.79 (raw)
Bovine-LTID Pipeline (Ours)	99.33	10.80	8.80	91.02 (GLTC)	60.00	1.66 (raw)

LLTC = Local Long-Term Coverage; GLTC = Global Long-Term Coverage; MT = Mostly-Tracked targets ($\geq 80\%$). For Bovine-LTID, multi-camera fusion yields a relative reduction of 87.6% in raw switches (Cohen's $d = 1.66$) and 87.6% in semantic switches (Cohen's $d = 1.70$) relative to its single-view tracking inputs.

Global metrics definition. **Cross-camera consistency** measures the proportion of temporally aligned frames where an individual livestock instance simultaneously visible in both camera perspectives receives an identical global ID. **Global ID switch rate** (normalized per 1000 frames) quantifies the frequencies of identity transitions observed along the synchronized trajectories. To differentiate transient boundary handovers from repetitive trajectory oscillations, the **semantic ID switch rate** isolates $A \rightarrow B \rightarrow A$ identity patterns, counting each re-association instability as a single event. **Global Long-Term Coverage (GLTC)** extends view-specific metrics to a multi-view context, representing the frame-weighted mean temporal permanence of global identities, defined as $GLTC = \sum_g w_g c_g$. Concurrently, the **mostly-tracked (MT)** ratio denotes the percentage of individual global trajectories whose temporal coverage spans greater than or equal to 80% of their physiological presence.

Statistical effect size and matched tracking metrics. To establish a statistical analysis of tracking stabilization, we contrast the single-view baseline tracking inputs against the centralized cross-camera tracking outputs under matched measurement conventions. The tracking change magnitude is assessed via the relative percentage reduction in identity switches alongside Cohen's d as a standardized indicator of statistical effect size.

Quantitative tracking stability evaluations validated against manual ground truth are consolidated in Table 9. The empirical data indicate a decoupling between spatial frame-level pairing and long-term trajectory persistence within our architectural design. When evaluating the frame-level cross-camera identity alignment percentage, both the pure geometric reference pipeline (H+NN Baseline) and our multi-feature architecture achieve comparable spatial consistency scores, yielding 99.19% and 99.33% respectively. This minimal variation (+0.14 percentage points) indicates that ground-plane homography projection provides an effective spatial reference for cross-view target matching under standard line-of-sight configurations.

However, a performance divergence manifests in identity lifecycle retention and switch frequency metrics. In the local single-view tracking phase, the fine-tuned CowTracker module operates under conservative motion and scale gates, terminating trajectories during high-density occlusion boundaries to prevent false-positive tracking

linkages. This design condition elevates the single-camera local tracklet switch rates to 100.70 (Camera 1) and 73.80 (Camera 2) per 1000 frames, generating clean but fragmented tracking data.

When entering the central cross-camera synchronization phase, the H+NN Baseline exhibits limitations in managing these local trajectory fragmentations. Bypassing global memory structures, the basic nearest-neighbor matching propagates localized tracking updates directly, leading to localized tracking cross-contamination. This vulnerability is reflected by a global raw identity switch rate of 53.90 per 1000 frames, which exceeds its view-specific inputs, resulting in a net increase in global identity switches relative to its single-view inputs (Cohen's $d = -0.79$).

In contrast, the complete Bovine-LTID pipeline utilizes its centralized GIDP layers and over-limit buffering to reconcile these view-specific tracking fragmentations. By suspending tracklet states during transient occlusions and smoothing coordinate jitter via spatial topology constraints, Bovine-LTID reduces global tracking switches to a raw rate of 10.80 and a semantic rate of 8.80 per 1000 frames. This corresponds to an 87.6% reduction in raw identity switches relative to its per-camera tracking averages ($87.2 \rightarrow 10.8$). The corresponding standardized effect size (Cohen's $d = 1.66$ for raw switches; $d = 1.70$ for semantic switches) indicates a measurable extension in global identity continuity. This minimization of tracking switches expands the Global Long-Term Coverage (GLTC) to 91.02%, representing a trajectory baseline suitable for continuous individual behavioral analytics.

As established in Section 4.2, the absolute metrics across the ground-truth validation subset (Table 9) and the full unannotated proxy log (Table 10) are not directly comparable due to distinct identity groundings and measurement denominators. However, relative trajectory tracking trends remain consistent within each independent paradigm. As shown in Table 10, when extended over the full 10-minute unannotated sequence, Bovine-LTID reduces the semantic global identity switches to 12.70 per 1000 frames. Compared against its raw localized tracking inputs, this cross-camera synchronization yields an 87.1% relative reduction in identity oscillations with a notable statistical effect size (Cohen's $d = 2.26$). As visually summarized in Fig. 12(a), the transition from single-view tracks to multi-view global

Table 10

Proxy cross-camera tracking stability evaluation executed on the complete 10-minute unannotated sequence. These indicators are derived from spatio-temporal self-consistency criteria and describe system baseline coherence over extended operational durations.

Configuration	X-cam Consist. (proxy, %) ↑	Raw Sw/1k (proxy) ↓	Semantic Sw/1k (proxy) ↓	GLTC (proxy, %) ↑	MT (proxy, %) ↑
Per-camera (Cam 1, w/ recovery)	–	108.80	–	19.54 (LLTC)	–
Per-camera (Cam 2, w/ recovery)	–	87.80	–	17.40 (LLTC)	–
Cross-camera (Bovine-LTID)	98.62	35.30	12.70	91.02	60.00

Proxy metrics defined in Appendix E. For Bovine-LTID, global fusion relative to single-view tracking inputs yields a 64.1% reduction in raw global switches (Cohen's $d = 1.66$) and an 87.1% reduction in semantic switches (Cohen's $d = 2.26$).

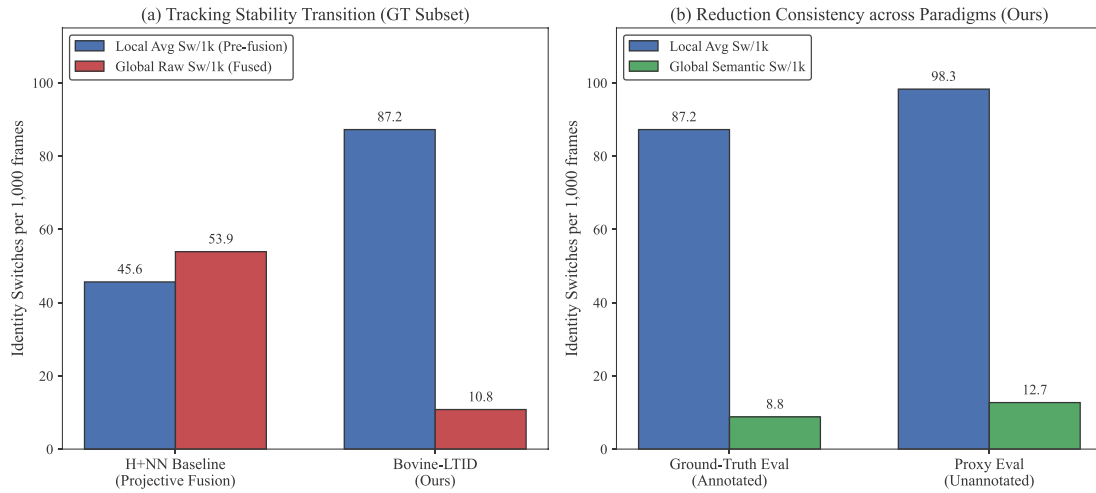


Fig. 12. Visual analysis of tracking stabilization with independent 0–100 metrics scales. (a) Transitioning from single-view tracks to cross-view fusion corresponds with an increase in identity switches for the H+NN Baseline, whereas Bovine-LTID maintains trajectory stability. (b) Despite differing absolute values due to independent measurement frameworks (GT vs. Proxy), the trend of tracking switch reduction remains consistent for Bovine-LTID.

trajectories highlights the limitations of the H+NN Baseline, which experiences an increase in identity switches. Conversely, Bovine-LTID effectively fuses tracklets to stabilize identities. Furthermore, Fig. 12(b) confirms that this trend—the reduction of identity switches following cross-camera fusion—is consistent across both the GT and Proxy evaluation paradigms, indicating the stability of the proposed temporal architecture without relying on appearance ReID.

This quantitative trajectory stability translates into localized visual tracking continuity. As demonstrated in the visual comparison captured at frame 6000 of the sequence (Fig. 13), the proposed Bovine-LTID pipeline handles congested configurations within the feeding troughs. During prolonged mutual occlusions and orientation variations where geometric coordinate matching experiences tracking interruptions, our spatio-temporal framework maintains consistent individual identification overlays, providing a dependable data stream for downstream agricultural informatics.

4.4. Long-term individual-level behavioral analytics

With globally consistent identities preserved across long recordings, the Bovine-LTID pipeline bridges low-level tracking metrics to high-level agricultural informatics.

Fig. 14 presents the quantitative analytics extracted directly from the system's global trajectory logs: detailed movement patterns via spatial occupancy heatmaps, ethological timelines mapping continuous state transitions, and time-budget distributions aggregating activities into actionable summaries. These high-fidelity, data-driven individual profiles empower farm managers to visually and quantitatively identify deviations from normal diurnal rhythms—for instance, altered lying duration and resting patterns as early indicators of lameness (von Keyserlingk et al., 2012), or broader behavioral deviations suggesting underlying health or welfare issues (Weary et al., 2009).

We emphasize that these behavioral-clinical associations are derived from established veterinary literature and that Bovine-LTID is

designed to function as an automated pre-screening tool rather than a definitive diagnostic instrument. Consequently, we recommend that any systemic behavioral alerts be subjected to expert veterinary review prior to clinical intervention. By translating complex spatio-temporal data into accessible, rigorous quantitative evidence, Bovine-LTID supports the transition from reactive treatment to proactive, individual-level herd care, a core objective of AI-powered e-monitoring (Moe et al., 2025). To demonstrate the system's robustness across higher herd densities, an extended 5-cow quantitative analysis over a 10-minute sequence is provided in Appendix H.

5. Discussion

As established in the preceding sections, this study proposes a spatio-temporal, ReID-embedding-free approach tailored for complex barn environments. While single-camera methodologies (Myat Noe et al., 2025; Li et al., 2022) often suffer from identity fragmentation due to dense occlusions in feeding zones, our Bovine-LTID addresses these physical constraints through homography-based multi-view fusion.

By integrating the cattle-specific CowTracker with a Global ID Pool, our method achieves a 99.33% cross-camera consistency and reduces semantic identity switches by 87.6% over a 10-minute sequence. Unlike traditional general-purpose trackers (e.g., DeepSORT (Wojke et al., 2017) and Deep OC-SORT (Maggiolino et al., 2023)) that struggle under prolonged occlusions, our framework provides a resource-efficient approach for maintaining individual identities without visual ReID.

Scalability analysis and roadmap. A key consideration for translating the current proof-of-concept to practical deployment is how the system's core components scale with increasing herd size. We therefore provide a qualitative complexity analysis and propose concrete directions for future empirical validation.

At the single-camera level, CowTracker's per-frame complexity is dominated by the multi-constraint scoring step, which scales as $O(|T| \cdot |D|)$ where $|T|$ is the number of active tracks and $|D|$ the number of

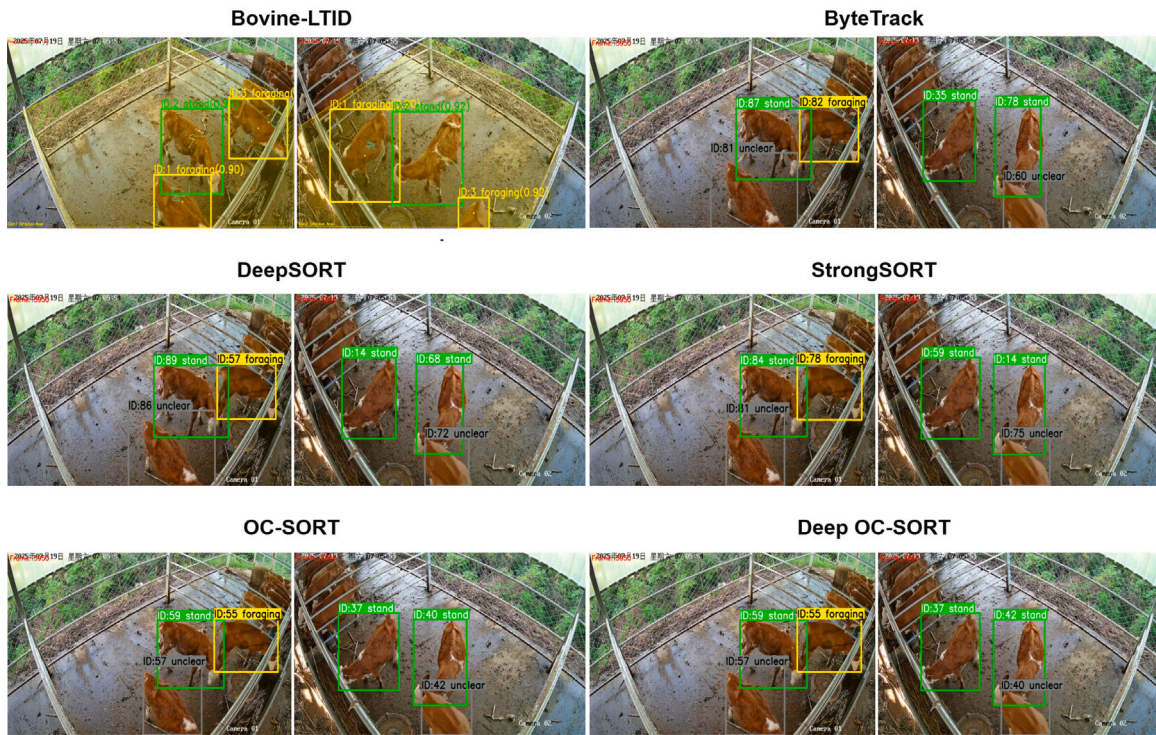


Fig. 13. Visual comparison of tracking performance at frame 6000 of the 10-minute sequence, illustrating consistent cross-view individual identification.

detections. With typical barn densities, both quantities are expected to grow approximately linearly with the herd size N , yielding an overall complexity of $O(N^2)$ per frame. The ID-stability protection (E3) and trajectory continuity verification (E2) operate on individual tracks and add only constant-factor overhead. The dynamic ID ceiling mechanism (E4) scales as $O(N)$ and is not expected to be a bottleneck.

At the cross-camera level, the Hungarian-algorithm-based identity matching has $O(N^3)$ worst-case complexity. While this is manageable for the small-group setting evaluated in this study ($N \leq 5$), it is expected to become the dominant scaling bottleneck as herd size grows. For larger herds, the cost matrix expands quadratically and the Hungarian solver's cubic complexity may approach the limits of real-time computation, depending on the target frame rate and available hardware.

We emphasize that the above analysis is based on algorithmic complexity rather than empirical profiling; systematic benchmarking across herd sizes is planned as future work. To guide this effort, we identify several complementary strategies that may mitigate the identified bottlenecks. First, spatially partitioned matching can divide the barn into functional zones, reducing the effective N per cross-camera solve by restricting association to cattle within the overlapping field of view. Second, GPU-accelerated linear assignment (e.g., batch Hungarian or Sinkhorn-based differentiable assignment) can exploit the parallelism of modern GPUs for larger cost matrices. Third, gated matching with a loose spatial pre-filter (e.g., world-coordinate proximity within approximately 2 m) can prune the majority of candidate pairs before the Hungarian step, yielding near-linear complexity in practice. These strategies form a concrete roadmap for future work aimed at extending Bovine-LTID to larger commercial herds.

Despite achieving promising results in tracking and behavior recognition, this study still presents certain limitations.

First, the data used in this study featured a relatively small sample size, tracking up to five cattle simultaneously within the overlapping field of view under controlled camera configurations. This limits the immediate generalizability of the findings to larger herds in commercial feedlot settings. We emphasize that the present work serves as a proof-of-concept demonstration of the ReID-embedding-free approach, and

the scaling roadmap outlined above provides a concrete pathway for extending the system to higher-density environments.

Second, all experiments were conducted in a single barn with Simmental cattle, and the detection model was trained exclusively on this breed. This limits the evidence for cross-breed and cross-environment generalizability. We note, however, that several core design aspects are inherently breed- and environment-agnostic: the homography-based coordinate unification relies only on geometric calibration and does not depend on visual coat patterns; the spatio-temporal logic of CowTracker (multi-constraint scoring, trajectory continuity, and ID-stability protection) operates on motion and size cues that are species-level rather than breed-level; and the Global ID Pool management is independent of cattle appearance. Conversely, the behavior-aware detector has been fine-tuned on Simmental-specific appearance features, and its performance on visually distinct breeds with different body conformations and coat textures remains unvalidated. The dynamic distance thresholds in CowTracker are parameterized for the body dimensions of the studied breed; morphologically distinct breeds may require threshold recalibration. The barn-specific camera geometry, lighting conditions, and fixed physical marker positions further constrain the direct transferability of the current trained model to new environments without per-site fine-tuning.

Third, although the proposed fusion mechanism substantially reduces identity switches, it does not achieve 100% elimination. Occasional identity fragmentation may still occur under prolonged mutual occlusions. Consequently, for autonomous engineering deployments in commercial farms that demand low error tolerances, the system may still require periodic manual trajectory verification.

Fourth, the homography-based coordinate alignment assumes a predominantly planar ground surface. To assess the practical impact without returning to the barn, we performed a synthetic geometric sensitivity analysis. The monitored area spans $4.0 \text{ m} \times 3.5 \text{ m}$, with the farthest point approximately 2.5 m from the calibration-plane center. Under a floor slope of θ , the vertical projection error at this distance is approximately $2.5 \cdot \tan \theta$. For a typical barn floor slope of 2° , the error is $\approx 9 \text{ cm}$; even at a pronounced 5° slope, it reaches $\approx 22 \text{ cm}$. Both values

Individual-Level Behavioral Analytics — Bovine-LTID Output

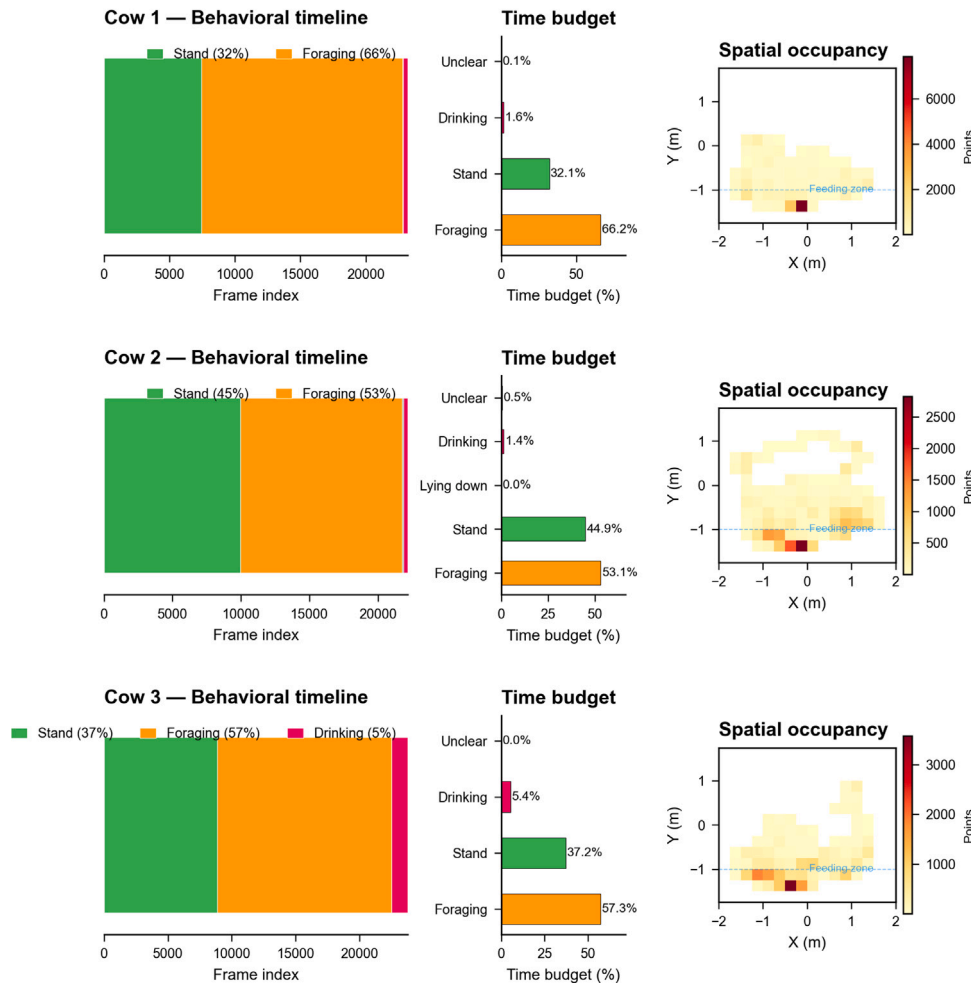


Fig. 14. Quantitative individual-level behavioral analytics generated from Bovine-LTID tracking logs over a 40-minute sequence. **Left:** Continuous behavioral state timelines (ethograms) demonstrating long-term identity persistence for three cows. **Middle:** Aggregated time-budget distributions quantifying resource utilization. **Right:** World-coordinate spatial occupancy heatmaps confirming targeted physical movement (e.g., concentrated activity within the feeding zone).

lie comfortably within the cross-camera Y-coordinate matching threshold of 0.6–0.8 m (Section 3.3), indicating that the current matching logic has inherent tolerance to moderate non-planarity. Heavily littered surfaces introduce transient, localized deviations rather than systematic biases and are partially absorbed by the exponential moving average smoothing applied to track coordinates before projection. We note that this analysis is geometric and does not substitute for empirical validation; systematic in-situ measurements with controlled slope variations are planned as future work.

Fifth, while Bovine-LTID presents a purely kinematic, ReID-free alternative to recent multi-modal tracking approaches, a direct quantitative comparison against contemporary state-of-the-art frameworks relying on periodic face recognition (e.g., Han et al. (2025)) was precluded by strict hardware disparities (i.e., the absence of frontal face cameras in our top-down setup) and private dataset constraints. Establishing standardized, multi-modal benchmarks to empirically compare these divergent architectural philosophies remains an important direction for future comparative research.

Future work will aim to expand and deepen this research in the following aspects. First, the sample size and data sources will be increased

to validate the system on larger herds across diverse farming environments and multiple cattle breeds, thereby improving the generalizability of both the detection model and the tracking heuristics. Second, we will explore online homography refinement and depth-aware projection techniques (Dickson et al., 2024) to enhance cross-view projection robustness in non-planar settings, alongside the potential integration of this system with autonomous robotic platforms for extended barn coverage. Third, following the scaling roadmap outlined above, we plan to implement and benchmark spatially partitioned matching and GPU-accelerated assignment to quantify the practical scalability ceiling of the framework. Fourth, to address the single-barn limitation, we will conduct multi-site transfer experiments quantifying how much per-site fine-tuning data is required for the detection module and whether the spatio-temporal tracking heuristics generalize with minimal recalibration. Furthermore, while the proposed manual spatial heuristics proved effective and stable for confined indoor barns, developing adaptive spatial learning methods remains a promising direction for tracking in unconstrained, variable environments such as outdoor pastures. Finally, we plan to leverage advanced machine learning algorithms to analyze the long-term observational data generated by our system, transforming

raw behavioral trajectories into automated, predictive insights for early disease warning and personalized herd management.

6. Conclusion

We presented Bovine-LTID, a lightweight, ReID-embedding-free, multi-camera tracking and behavioral analysis system for indoor barn environments housing small groups of cattle. By combining a behavior-aware YOLOv9s detector (yielding an overall test-set mAP50 of 0.944 ± 0.003 and mAP50-95 of 0.885 ± 0.008), the cattle-specific CowTracker, periodic EdgeSAM refinement, and homography-based coordinate unification, the system addresses the persistent challenge of identity fragmentation. On a 10-minute manually annotated dual-camera monitoring sequence, our Global ID Pool and cross-view behavior inference mechanism achieve a cross-camera identity consistency of 99.33% and reduce global raw identity switches by 87.6% relative to single-camera tracking inputs, with 60% of the herd classified as mostly-tracked (Cohen's $d = 1.66$).

We have further provided a systematic scalability analysis identifying the key computational bottlenecks and outlining concrete optimization strategies—including spatially partitioned matching and GPU-accelerated assignment—as a roadmap toward extending the system to larger herds.

The results obtained at the current proof-of-concept scale suggest that spatio-temporal consistency, rather than appearance-based ReID, can serve as a viable alternative for identity preservation in multi-camera cattle monitoring. While further validation on larger and more diverse herds is needed, this work provides a baseline framework for resource-efficient, ReID-free approaches in precision livestock farming.

CRedit authorship contribution statement

Boyang Zhao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Bo Xiao:** Validation, Resources. **Cong Liu:** Validation, Resources. **Xiao Jin:** Validation, Resources. **Jun-Ming Ye:** Validation, Data curation. **Mingjie Yu:** Validation, Data curation. **Zhenhua Yu:** Writing – review & editing, Supervision, Resources, Project administration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

A.1. Practical deployment and maintenance checklist

To ensure the long-term robustness of the Bovine-LTID homography calibration in dynamic barn environments, the following operational checklist is recommended for practical deployment:

- **1. Hardware Fixation:** Ensure all ceiling camera mounts are rigidly bolted to structural beams to minimize vibrations from farm machinery.
- **2. Permanent Anchoring:** Mark the four calibration reference points (forming the 4.0 m $\times$ 3.5 m area) using highly durable, wear-resistant materials (e.g., deeply embedded markers or epoxy paint) to withstand cleaning and animal hoof wear.

Table B.11

Hyperparameter sensitivity analysis of the enhanced multi-feature matching score distributions. The table presents the statistical distribution of assignment scores under varying weight configurations.

Configuration	Mean	Std	Min	Max
<i>Base Similarity Weight (w_b) Sweep</i>				
$w_b = 0.3$	0.6234	0.2235	0.3065	0.9151
$w_b = 0.4$	0.6588	0.1979	0.3753	0.9208
$w_b = 0.5$ (Default)	0.6942	0.1727	0.4442	0.9264
$w_b = 0.6$	0.7296	0.1480	0.5130	0.9321
$w_b = 0.7$	0.7650	0.1240	0.5819	0.9378
<i>Relative Position Weight (w_r) Sweep</i>				
$w_r = 0.2$	0.7530	0.1081	0.5932	0.9042
$w_r = 0.3$	0.7218	0.1396	0.5173	0.9131
$w_r = 0.4$ (Default)	0.6942	0.1727	0.4442	0.9264
$w_r = 0.5$	0.6666	0.2063	0.3711	0.9398
$w_r = 0.6$	0.6353	0.2398	0.2952	0.9487
<i>Trajectory Weight (w_t) Sweep</i>				
$w_t = 0.00$	0.7173	0.1905	0.4415	0.9736
$w_t = 0.05$	0.7058	0.1816	0.4428	0.9500
$w_t = 0.10$ (Default)	0.6942	0.1727	0.4442	0.9264
$w_t = 0.15$	0.6826	0.1639	0.4455	0.9028
$w_t = 0.20$	0.6711	0.1550	0.4469	0.8792

- **3. Software Initialization:** Map the pixel coordinates of the anchors to the hardcoded world coordinates to establish the baseline homography matrix.
- **4. Periodic Verification (Monthly):** Visually inspect the camera feed to verify that the hardcoded software ROI boundaries still perfectly align with the physical floor markers.
- **5. Condition-Based Recalibration:** Only update the hardcoded coordinates if a deliberate camera repositioning occurs or if a severe misalignment is detected during the periodic verification.

Appendix B. Hyperparameter sensitivity analysis of matching weights

To further assess the robustness of the multi-feature matching algorithm presented in Section 3.3, a sensitivity analysis was conducted on the cross-camera assignment weights: Base Similarity (w_b), Relative Position Consistency (w_r), and Trajectory Consistency (w_t). Table B.11 details the statistical distribution (Mean, Standard Deviation, Minimum, and Maximum) of the enhanced match scores under varying configurations. The analysis confirms that the selected default configuration ($w_b = 0.5$, $w_r = 0.4$, $w_t = 0.1$) provides a balanced score distribution, preventing any single feature term from disproportionately dominating the assignment matrix.

Appendix C. EdgeSAM interval sensitivity analysis

To empirically justify the every-3-frames EdgeSAM execution setting described in Section 2.2, a controlled sensitivity experiment was conducted on the 3000-frame dual-camera ground-truth subset. Five interval configurations were tested: executing the segmentation mask refinement every 1, 2, 3, 5, and 10 frames.

The empirical results (Table C.12) indicate that every evaluated tracking metric remained effectively constant across all five interval settings: F1 and IDF1 varied by < 0.001 , cross-camera consistency by $< 0.1\%$, and the semantic switch rate by < 0.1 per 1000 frames. Only the end-to-end processing throughput varied meaningfully, increasing from 7.2 FPS with per-frame SAM execution to 16.9 FPS at an every-10-frames interval.

Two primary conclusions can be drawn from this analysis. First, the robustness of CowTracker is governed by its spatio-temporal logic rather than reliance on high-frequency segmentation quality, which

Table C.12

EdgeSAM interval sensitivity analysis with ground-truth validation. Local single-view metrics are presented as Camera 1/Camera 2, alongside aggregate global metrics.

Int.	Local (Cam1/Cam2)				Global			
	F1	IDF1	ID Sw/1k	LLTC %	X-cam %	Sem Sw/1k	GLTC %	FPS
1	0.993/.995	0.759/.774	120.7/37.0	51.6/55.1	99.65	5.7	98.17	7.2
2	0.993/.995	0.759/.774	120.7/37.0	51.6/55.1	99.65	5.7	98.17	11.8
3	0.993/.995	0.759/.774	120.7/37.0	51.6/55.1	99.65	5.7	98.17	13.6
5	0.993/.995	0.759/.774	120.7/37.0	51.6/55.1	99.65	5.7	98.17	15.4
10	0.993/.995	0.759/.774	120.7/37.0	51.6/55.1	99.65	5.7	98.17	16.9

Table D.13

Detailed hyperparameter configurations for CowTracker, Multi-Camera Fusion, and EdgeSAM.

Parameter description	Value/formula
CowTracker: Multi-Constraint Scoring (E1)	
Robust matching weights ($\lambda_1, \lambda_2, \lambda_3$)	0.3, 0.3, 0.4
Fallback matching weights (γ_1, γ_2)	0.5, 0.5
Dynamic distance threshold (D_{dyn}) formula	$\text{clip}(M_{\text{traj}} \sqrt{\text{area}/10^5} \cdot 30/\text{fps}, 10, 50)$
Position, Size, Direction weights (w_p, w_s, w_d)	0.7, 0.1, 0.2
CowTracker: Trajectory & Stability Verification (E2, E3)	
Base jump limit (M_{jump})	60 pixels
Jump threshold $D_{\text{jump}}^{\text{th}}(\Delta t)$	$10M_{\text{jump}} (\Delta t > 5)$ $5M_{\text{jump}} (2 < \Delta t \leq 5)$ $(1 + 0.5\Delta t)M_{\text{jump}} (\Delta t \leq 2)$
Maximum velocity bounds (V_{max}, m)	60 px/frame, $m = 2.0$
Switch rejection window ($W_{\text{frames}}, N_{\text{switch}}$)	10 frames, ≥ 3 switches
Over-limit buffer duration (F_{buffer})	10 frames
Bovine-LTID: Cross-Camera Fusion	
Maximum spatial association distance (d_{max})	0.5 meters
Initial minimum similarity threshold (τ_{sim})	0.6
Multi-feature weights (w_b, w_r, w_i)	0.5, 0.4, 0.1
Base similarity sub-weights (w_d, w_s, w_c)	0.7, 0.2, 0.1
Y-coordinate mismatch threshold (ΔY_{max})	0.6 ~ 0.8 meters
Trajectory history window (L)	5 frames
Re-association acceptance threshold (τ_{reassoc})	0.6
GIDP track stability requirement (F_{stable})	100 frames
Mask consistency contribution weight (w_m)	0.25
Sparse visualization displacement threshold	0.2 meters
Maximum physical jump filter (artifact rejection)	2.0 meters
Bounded deque memory limit (L_{queue})	50 elements
EdgeSAM Module	
Frame interval	3 frames
Minimum detection confidence	0.5

is consistent with the ReID-embedding-free design philosophy. Second, the every-3-frames default (~13.6 FPS) represents a well-balanced engineering choice, delivering more than double the throughput of per-frame execution without incurring any measurable penalty to tracking continuity.

(Note: The higher local ID switch rates in Table C.12 compared to the baseline single-camera benchmarks reflect a distinct experimental configuration where the ID recovery module is fully enabled, proactively re-assigning identities post-occlusion. As evidenced by the stable global metrics, these localized switches are successfully resolved by the cross-camera fusion layer.)

Appendix D. Implementation details and hyperparameters

See Table D.13.

Appendix E. Proxy tracking metrics

For long-term video sequences where exhaustive manual identity annotation is impractical, we propose a proxy evaluation framework that assesses tracking quality through spatio-temporal self-consistency

rather than external ground-truth labels. This appendix provides the complete mathematical definitions.

E.0.1. Physical track construction

Local physical tracks. Given the local tracking log $\{(f, c, \ell, x_1, y_1, x_2, y_2)\}$ where f is the frame index, c the camera, ℓ the system-assigned local track ID, and (x_1, y_1, x_2, y_2) the bounding box, we construct *physical tracks* by greedily matching detections across consecutive frames via Hungarian assignment at $\text{IoU} \geq \tau_{\text{IoU}} = 0.7$ within a maximal time gap $\Delta t_{\text{max}} = 3$ s. Each physical track P_k is an ordered sequence of (f, ℓ, b) tuples.

Global physical tracks. From the global tracking log $\{(f, c, g, X, Y)\}$ where g is the global ID and (X, Y) the world coordinate, we construct physical tracks analogously using world-coordinate Euclidean distance at threshold $d_{\text{world}} = 0.5$ m.

E.0.2. Single-camera proxy metrics

ID switch rate. (per 1000 frames). Within each physical track P_k , an ID switch occurs whenever the assigned local ID ℓ changes between consecutive frames. The rate is $R_{\text{switch}} = 1000 \cdot N_{\text{switches}}/T$, where T is the total number of frames.

Proxy IDF1. For each physical track P_k , let $m_{\text{dom}}^{(k)}$ be the count of the most frequent (dominant) local track ID appearing in that track. Let D_{phys} be the total number of detections covered by all physical tracks, and D_{loc} the total detections in the raw local tracking log. Then $\text{Proxy IDF1} = 2 \sum_k m_{\text{dom}}^{(k)} / (|D_{\text{phys}}| + |D_{\text{loc}}|)$.

Weighted local long-term coverage (LLTC). For each local track ID i , let n_i be the number of frames in which i appears, $c_i = n_i/T$ its temporal coverage, and $w_i = n_i / \sum_j n_j$ its frame-weighted proportion. Then $\text{LLTC} = \sum_i w_i c_i$.

Long-track coverage. (≥ 30 s). Let S be the set of contiguous same-ID segments within all physical tracks, and d_s the duration (in seconds) of segment s . Then $\text{Long-track ratio} = \sum_{s: d_s \geq 30s} d_s / \sum_s d_s$.

Occlusion recovery rate. For each physical track P_k , successive frame entries separated by a gap > 0.2 s define adjacent segments. The occlusion recovery rate is the fraction of adjacent segment pairs where the dominant local ID is the same: $N_{\text{same-dom}} / N_{\text{adjacent-pairs}}$.

E.0.3. Cross-camera proxy metrics

Cross-camera consistency. For each frame where a physical target appears in both camera views, we check whether the system assigns consistent global IDs across the two cameras. The consistency is the proportion of such overlapping-frame instances where both cameras agree on the global identity.

Semantic global ID switch. (A $\rightarrow$ B $\rightarrow$ A counted once). For each global physical track P_k , we extract the sequence of distinct global IDs in chronological order (merging consecutive duplicates). A *semantic switch* is counted when a previously observed global ID reappears after a different global ID was assigned (A $\rightarrow$ B $\rightarrow$ A). Simple transitions (A $\rightarrow$ B $\rightarrow$ C, where each ID is new) are counted in the *raw* switch count but excluded from the semantic count. This distinction ensures that transient identity oscillations are counted once rather than inflated by repeated back-and-forth transitions.

Raw global ID switch. Every transition between distinct global IDs along a physical trajectory is counted once, regardless of whether the target ID has been observed before. This definition is directly comparable to the per-camera local ID switch rate (both count every frame-to-frame identity change), enabling effect-size computation with matched denominators (Section 4.3).

Global long-term coverage (GLTC). For each global ID g , let n_g be the number of frames in which g appears across either camera, $c_g = n_g/T$, and $w_g = n_g / \sum_{g'} n_{g'}$. Then $\text{GLTC} = \sum_g w_g c_g$. The *mostly-tracked ratio* is the fraction of global IDs with $c_g \geq 0.8$.

Effect size. To compute the reduction in ID switches with a matched denominator, we average the per-camera raw switch rates $\bar{R}_{\text{local}}$ and compare against the global raw switch rate R_{global} . $\text{Reduction}_{\text{raw}} = (\bar{R}_{\text{local}} - R_{\text{global}}) / \bar{R}_{\text{local}}$. Cohen's d is computed as $d = (\bar{R}_{\text{local}} - R_{\text{global}}) / s_{\text{pooled}}$, where s_{pooled} is the pooled standard deviation of the two per-camera rates and the global rate.

Appendix F. CowTracker pseudocode

The complete cascaded identity recovery and track management logic of the CowTracker module, highlighting the integration of the four core algorithmic enhancements (E1–E4), is formalized in Algorithm 1.

Appendix G. Cross-camera fusion pseudocode

The step-by-step logic of the centralized multi-camera synchronization phase, integrating both matching stages (world-distance and multi-feature) alongside the Global ID Pool (GIDP) stability gates, is detailed in Algorithm 2.

Appendix H. Extended quantitative behavioral analytics (higher-density scenario)

To further validate the scalability and robustness of the Bovine-LTID system in a higher-density scenario, Fig. H.15 presents the quantitative behavioral analytics for a 5-cow group over a 10-minute (6000-frame) monitoring sequence. The system successfully extracts continuous ethograms, time-budget distributions, and spatial occupancy heatmaps for all five individuals without identity loss, demonstrating its capability to handle increased mutual occlusions and complex multi-agent interactions within the feeding zone.

Algorithm 1 Cascaded ID Recovery and Track Management (CowTracker)

Require: YOLO detections D_t (boxes, classes, scores), active tracks $\mathcal{T}_{t-1}$ with history, cross-camera ceiling N_{cross}

Ensure: Updated tracks $\mathcal{T}_t$ with assigned cow_id

```

1: // Step 0: ByteTrack first-pass
2: Run ByteTrack association on  $D_t \rightarrow \mathcal{O}_t$ 
3:  $N_{\text{current}} = |\mathcal{O}_t|$ ;  $N_{\text{ceiling}} = \max(N_{\text{current}}, N_{\text{cross}})$ 
4: // Step 1: Multi-constraint scoring (E1)
5: for each track  $o \in \mathcal{O}_t$  do
6:   Compute  $S_{\text{match}}(o, d_j)$  for each  $d_j \in D_t$ 
7:    $d^* = \arg \max S_{\text{match}}(o, d_j)$ 
8:    $o.\text{tlbr} \leftarrow d^*.\text{box}$ ,  $o.\text{class} \leftarrow d^*.\text{class}$ 
9: // Step 2: Cascaded ID recovery
10: for each strategy in cascade order do
11:   Propose candidate ID  $\text{cid}$ 
12:   if  $\text{cid}.\text{num} > N_{\text{ceiling}} \wedge \text{cid} \notin B_{\text{over}}$  then Reject (E4)
13:   if  $\neg \text{CHECKTRAJECTORY}(\text{cid}, d^*)$  then Reject (E2)
14:   if  $\neg \text{CHECKSTABILITY}(\text{cid})$  then Reject (E3)
15:   Accept  $\text{cid}$ ;  $o.\text{cow\_id} \leftarrow \text{cid}$ ; break
16: end for
17: Update stability & apply smoothing if stable  $\geq 5f$ 
18: Append to track history (trajectory, size, vel)
19: end for

18: // Step 3: Unmatched detections
19: for remaining  $d_k \in D_t$ , not matched do
20:   Run ID recovery cascade: (history  $\rightarrow$  inactive  $\rightarrow$  new-ID)
21:   Apply E2/E3/E4 gates before assignment
22:   if ID assigned then
23:     Create/Update track with recovered  $\text{cid}$ 
24:   end if
25: end for

26: // Step 4: Housekeeping
27: Update  $B_{\text{over}}$ : expire IDs older than  $F_{\text{buffer}}$  frames
28: Prune tracks inactive  $> 150$  frames; compact ID space
29: return  $\mathcal{T}_t$ 

```

Algorithm 2 Cross-Camera Bovine-LTID Identity Fusion (per-frame)**Input:** synchronized frame pair (F_1^1, F_2^2) , CowTracker instances $\{C_1, C_2\}$, H_1, H_2 **Global:** $\mathcal{M}_1, \mathcal{M}_2$ (re-ID $\rightarrow$ GID maps), N_{ceiling} (ID upper bound, hold-protected)**Output:** fused trajectories $\mathcal{G}$, cross-view behavior labels b

```

1: // Stage 1: Per-camera tracking (with sync)
2:  $C_1.\text{set\_ceiling}(N_{\text{ceiling}})$ ,  $C_2.\text{set\_ceiling}(N_{\text{ceiling}})$ 
3:  $\mathcal{T}_1^1 \leftarrow C_1.\text{update}(D_1^1)$ ,  $\mathcal{T}_1^2 \leftarrow C_2.\text{update}(D_1^2)$ 

4: // Stage 2: World coordinate projection
5: for each track  $i \in \mathcal{T}_1^1 \cup \mathcal{T}_1^2$  do
6:    $\mathbf{c}_i \leftarrow H_{\text{cam}(i)} \cdot (u_i, v_i, 1)^\top$ 
7: end for

8: // Stage 3: World-distance bipartite matching
9:  $\mathbf{C} \leftarrow \infty_{|D_1^1| \times |D_2^2|}$                                 cost matrix
10: for each box pair  $(i, j)$  do
11:   if  $\|\mathbf{c}_i^1 - \mathbf{c}_j^2\| \leq d_{\text{max}}$  then
12:      $\mathbf{C}[i, j] \leftarrow \|\mathbf{c}_i^1 - \mathbf{c}_j^2\|$ 
13:   end if
14: end for
15:  $(i^*, j^*) \leftarrow \text{Hungarian}(\min \sum \mathbf{C}[i, j])$ 
16:  $S_{\text{match}}[k] \leftarrow 1 - \mathbf{C}[i_k^*, j_k^*]/d_{\text{max}}$ 

17: // Stage 4: GIDP — matched-pair resolution
18:  $\text{ASSESSSTABILITY}(\mathcal{T}_1^1, \mathcal{T}_1^2)$ 
19: for each matched box pair  $(i^*, j^*)$  with  $S_{\text{match}} \geq \tau_S$  do
20:    $(r^1, r^2) \leftarrow \text{nearest-track}(\mathcal{T}_1^1[i^*], \mathcal{T}_1^2[j^*])$ 
21:    $\text{rid}_1 \leftarrow r^1.\text{cow\_id}$ ,  $\text{rid}_2 \leftarrow r^2.\text{cow\_id}$ 
22:   if  $\text{STABILITYPROTECT}(\text{rid}_1, \text{rid}_2)$  rejects then
23:     continue                                     (unstable side suppressed)
24:   end if
25:    $\text{gid} \leftarrow \text{RESOLVEGID}(\text{rid}_1, \text{rid}_2, \mathcal{M}_1, \mathcal{M}_2)$ 
26:   if  $\text{gid} = \text{None}$  then
27:      $\text{gid} \leftarrow \min\{k \notin \mathcal{G}_i^{\text{used}}\}$ 
28:   end if
29:    $\mathcal{M}_1[\text{rid}_1] \leftarrow \text{gid}$ ,  $\mathcal{M}_2[\text{rid}_2] \leftarrow \text{gid}$ 
30:    $\mathbf{c}_{\text{mid}} \leftarrow (\mathbf{c}_{r^1} + \mathbf{c}_{r^2})/2$ 
31:    $\mathcal{G}[\text{gid}].\text{matched}.\text{append}(\mathbf{c}_{\text{mid}})$ 
32: end for

34: // Stage 5: Enhanced matching for unmatched
35: for each unassigned track  $t \in \mathcal{T}_1^1 \cup \mathcal{T}_1^2$  do
36:   for each track  $t'$  on the other camera do
37:      $S \leftarrow w_b S_{\text{base}} + w_r S_{\text{relative}} + w_t S_{\text{traj}}$ 
38:      $\text{best} \leftarrow \arg \max S$ 
39:   end for
40:   if  $\text{best}.S \geq \tau_S$  and  $\text{best.gid} \notin \mathcal{G}_t^{\text{used}}$  then
41:      $\mathcal{M}_{\text{cam}(t)}[\text{rid}] \leftarrow \text{best.gid}$ 
42:   else
43:      $\text{gid} \leftarrow \min\{k \notin \mathcal{G}_t^{\text{used}}\}$ 
44:      $\mathcal{M}_{\text{cam}(t)}[\text{rid}] \leftarrow \text{gid}$ 
45:   end if
46: end for

47: // Stage 6: Trajectory update & jump filter
48: for each track  $t \in \mathcal{T}_1^1 \cup \mathcal{T}_1^2$  do
49:    $\text{gid} \leftarrow \mathcal{M}_{\text{cam}(t)}[t.\text{cow\_id}]$ 
50:    $\mathbf{c} \leftarrow H_{\text{cam}(t)} \cdot (t.\text{center})$ 
51:   if  $\|\mathbf{c} - \mathcal{G}[\text{gid}].\text{last}\| > \delta_{\text{jump}}$  then
52:     skip                                           (prevent history contamination)
53:   else
54:      $\mathcal{G}[\text{gid}].\text{cam}_{\text{cam}(t)}.\text{append}(\mathbf{c})$ 
55:   end if
56: end for
57:  $\text{PRUNEIDLE}(\mathcal{G})$                                      (idle > 30 frames)

58: // Stage 7: Cross-view behavior inference
59: for each  $\text{gid} \in \mathcal{G}_i$  do
60:    $c_{\text{enh}}^k \leftarrow (1 - w_m)c_{\text{YOLO}}^k + w_m \text{IoU}_{\text{mask}}$ 
61:    $(b_{\text{best}}, c_{\text{best}}) \leftarrow \text{SelectBest}(b^1, b^2, c_{\text{enh}}^1, c_{\text{enh}}^2)$ 
62:   if  $b_{\text{best}} = \text{unclear}$  and  $b_{\text{other}} \neq \text{unclear}$ :
63:     choose  $b_{\text{other}}$                                (unclear-suppressed)
64: end for

65: // Stage 8: ID ceiling hold mechanism
66:  $N_{\text{prop}} \leftarrow |\text{matches}| + \max(|\text{unm}^1|, |\text{unm}^2|)$ 
67: if  $N_{\text{prop}} > N_{\text{ceiling}}$  then
68:    $N_{\text{ceiling}} \leftarrow N_{\text{prop}}$ ;  $\text{hold\_cnt} \leftarrow 0$ 
69: else if  $N_{\text{prop}} < N_{\text{ceiling}}$  then
70:    $\text{hold\_cnt} \leftarrow \text{hold\_cnt} + 1$ 
71:   if  $\text{hold\_cnt} \geq 100$  then  $N_{\text{ceiling}} \leftarrow N_{\text{prop}}$ 
72: end if
73:  $\mathcal{G} \leftarrow \text{COMPACTIDS}(\mathcal{G}, N_{\text{ceiling}})$ 

74: return  $\mathcal{G}$  with dual-camera trajectories & behavior  $b$ 

```

[Auxiliary Logic: Stability-Priority GID Resolution]**Trigger:** One camera stable (≥ 100 frames, consistent count) while other experiences transient change.**Action:** When rid_1 and rid_2 map to conflicting GIDs, prefer the GID from the stable camera.**Rationale:** Prevents a temporary detection artifact on one view from corrupting long-standing identity on the other.**[Auxiliary Logic: ID Ceiling Hold ($H = 100$)]****Trigger:** $|D_i^k|$ temporarily drops due to occlusion or missed detection.**Action:** Defer N_{ceiling} reduction; keep higher bound for 100 frames before allowing descent.**Rationale:** A single-frame detection gap should not compress the ID space and force existing large-IDs to be overwritten.

Individual-Level Behavioral Analytics — Bovine-LTID Output

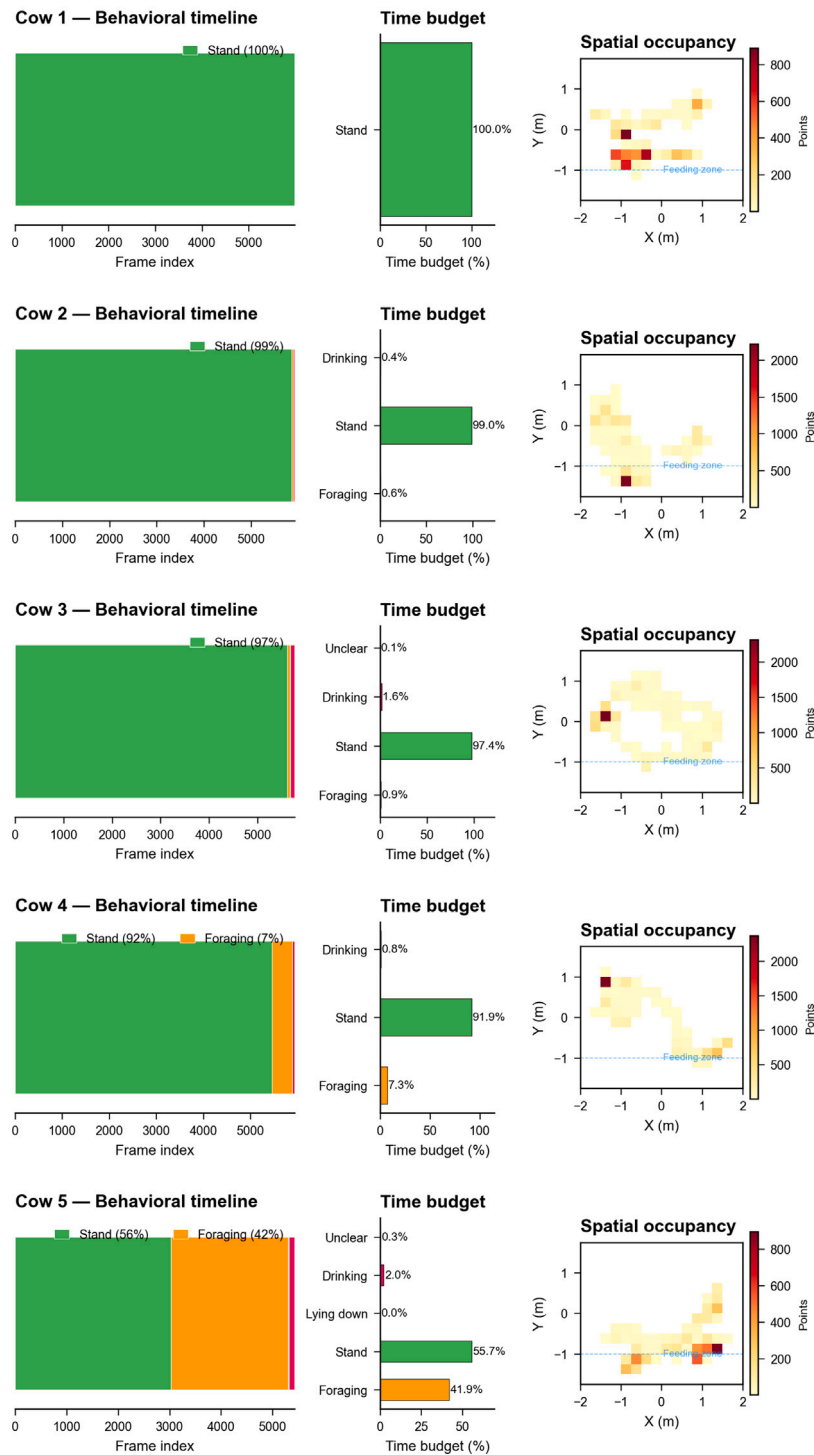


Fig. H.15. Extended quantitative behavioral analytics for a 5-cow group over a 10-minute monitoring sequence. The data confirms the system's ability to simultaneously generate high-fidelity behavioral profiles (timelines, time-budgets, and spatial heatmaps) for higher-density groups.

Data availability

Data will be made available on request.

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